Chapter 2 : Integration

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Chapter 2 : Integration

Exercise 2.1. Let $f : X \to \overline{\mathbb{R}}$ and $Y = f^{-1}(\mathbb{R})$. Then f is measurable iff $f^{-1}(\{-\infty\}) \in \mathcal{M}$, $f^{-1}(\{\infty\}) \in \mathcal{M}$, and $f|_Y$ is measurable on Y.

Solution. For the only if direction, $\{\infty\}, \{-\infty\}, \mathbb{R} = Y \in \mathcal{B}_{\overline{\mathbb{R}}}$ making f^{-1} map them into $\mathcal{M}$ as f is measurable and as $\mathcal{M}$ is a sigma algebra we see that $f|_Y : Y \to \overline{\mathbb{R}}$ is $\mathcal{M}_Y$-measurable as $f^{-1}(E) \cap Y \in \mathcal{M}_Y$ for all $E \in \mathcal{B}_{\overline{\mathbb{R}}}$. Now for the if direction, as $\mathcal{B}_{\overline{\mathbb{R}}} = \{E \cup I : E \in \mathcal{B}_{\mathbb{R}}, I \subseteq \{\infty, -\infty\}\}$ we see that for any such $E \cup I \in \mathcal{B}_{\overline{\mathbb{R}}}$ we have $f^{-1}(E \cup I) = f^{-1}(E) \cup f^{-1}(I)$ and as $f^{-1}(E) \cap Y = f^{-1}(E) \in \mathcal{M}_Y \subseteq \mathcal{M}$ and $f^{-1}(I) \in \mathcal{M}$ so f is measurable on $\mathcal{M}$. □

Exercise 2.2. Suppose $f, g : X \to \overline{\mathbb{R}}$ are measurable.

a. fg is measurable (where $0 \cdot (\pm\infty) = 0$).

b. Fix $\alpha \in \overline{\mathbb{R}}$ and define $h(x) = \alpha$ if $f(x) = -g(x) = \pm\infty$ and $h(x) = f(x) + g(x)$ otherwise. Then h is measurable.

Solution.

a. Consider the two maps,

$$X \ni x \longrightarrow (f(x), g(x)) \text{ and } \overline{\mathbb{R}}^2 \ni (x, y) \mapsto xy$$

the first is a measurable map from X (with sigma algebra $\mathcal{M}$) to $R \times R$ (with sigma algebra $\otimes_1^2 \mathcal{B}_{\overline{\mathbb{R}}}$) and the second from this to $\overline{\mathbb{R}}$ (with sigma algebra $\mathcal{B}_{\overline{\mathbb{R}}}$) and as the composition of measurable maps are measurable we are done. Here, the measurability of $(x, y) \to xy$ follows from its continuity (note: the metric on $\overline{\mathbb{R}}$ considered throughout the chapter is usually $d(x, y) := |\arctan(x) - \arctan(y)|$ with $\arctan(\pm\infty) = \pm\pi/2$).

b. Similarly again consider $x \mapsto (f(x), g(x))$ but this time with $(x, y) \to J(x, y)$ where,

$$J(x, y) := \begin{cases} \alpha \text{ if } x = -y = \pm\infty \\ x + y \text{ otherwise} \end{cases}$$

We will use Exercise 1.5 here. Let $A = \{(+\infty, -\infty), (-\infty, +\infty)\}$ and $B = \overline{\mathbb{R}}^2 \setminus A$, it is clear that J is continuous on B and hence measurable on it and as it is constant on A it is trivially measurable on it so we are done. As compositions of measurable functions are measurable we are done.

□

Exercise 2.3. If $\{f_n\}$ is a sequence of measurable functions on X, then $\{x : \lim f_n(x) \text{ exists}\}$ is a measurable set.

Solution. We know that $g_1 = \limsup f_n$ and $g_2 = -\liminf f_n$ are measurable functions, define

$$h(x) := \begin{cases} 2026 \text{ if } g_1(x) = -g_2(x) = \pm\infty \\ g_1(x) + g_2(x) \text{ otherwise} \end{cases}$$

h is a measurable function by Exercise 1.2 and f_n converges iff $g_1 = -g_2$ and we get that

$$\{x : \lim f_n(x) \text{ exists}\} = h^{-1}(\{0, 2026\})$$

which implies that the set infact is measurable as all the sets we performed unions and intersections on are in $\mathcal{M}$. □

Exercise 2.4. If $f : X \to \overline{\mathbb{R}}$ and $f^{-1}((r, \infty]) \in \mathcal{M}$ for each $r \in \mathbb{Q}$, then f is measurable.

Solution. It can easily be shown that these sets being in $\mathcal{M}$ implies that $f^{-1}((k, \infty]) \in \mathcal{M}$ for any $k \in \overline{\mathbb{R}}$ easily so we are done. □

Exercise 2.5. If $X = A \cup B$ where $A, B \in \mathcal{M}$, a function f on X is measurable iff $f|_A$ is measurable on A and $f|_B$ is measurable on B.

Solution. If f is measurable then measurability on $A, B \in \mathcal{M}$ are trivial. For the other direction, suppsoe $f : X \to Y$ with $\mathcal{N}$ being the sigma algebra on Y, for any $E \in \mathcal{N}$ we have,

$$f^{-1}(E) = (f^{-1}(E) \cap A) \cup (f^{-1}(E) \cap B) = (f|_A^{-1}(E)) \cup (f|_B^{-1}(E)) \in \mathcal{M}$$

as these two sets are in $\mathcal{M}_A, \mathcal{M}_B \subseteq \mathcal{M}$ respectively anyways. □

Exercise 2.6. The supremum of an uncountable family of measurable $\mathbb{R}$-valued functions on X can fail to be measurable (unless the σ-algebra $\mathcal{M}$ is very special).

Solution. Suppose $X = \mathbb{R}$ with some some sigma algebra containing singletons (most of the ones we care about do have them, for example the borel sigma algebra), then for any nonempty set $E \subseteq \mathbb{R}$ consider the family $\{\chi_{\{r\}}\}_{r \in E}$, these are all measurable as singletons are in the sigma algebra whereas $\sup_{r \in E} \chi_{\{r\}} = \chi_E$ always being measurable would imply that the sigma algebra was the whole powerset stemming from E being arbitrary. □

Exercise 2.7. Suppose that for each $\alpha \in \mathbb{R}$ we are given a set $E_\alpha \in \mathcal{M}$ such that $E_\alpha \subset E_\beta$ whenever $\alpha < \beta$, $\bigcup_{\alpha \in \mathbb{R}} E_\alpha = X$, and $\bigcap_{\alpha \in \mathbb{R}} E_\alpha = \varnothing$. Then there is a measurable function $f : X \to \mathbb{R}$ such that $f(x) \leq \alpha$ on E_α and $f(x) \geq \alpha$ on E_α^c for every α. (Use Exercise 4.)

Solution. We claim that the following function works

$$f(x) := \sup\{r \in \mathbb{Q} : x \notin E_r\}$$

For $x \in E_\alpha$, from the definition we have that $x \in E_\beta$ for all $\mathbb{Q} \ni \beta \geq \alpha$ and thus $f(x) = \sup\{r \in \mathbb{Q} : r < \alpha, x \notin E_r\} \leq \alpha$. Now, for $x \in E_\alpha^c$ its clearly $\geq \alpha$. We claim that, for $r \in \mathbb{Q}$ we have

$$f^{-1}((r, \infty)) = \bigcup_{\mathbb{Q} \ni \alpha > r} E_\alpha^c$$

If $f(x) > r$, then the set of β for which $x \in E_\beta^c$ (its an open ray unbounded below) must contain r, i.e its of form $(-\infty, v)$ with $v > r$ and we can find a rational $q \in (r, v)$ such that $x \in E_q$ so we have one direction of the equality, the other direction is trivial and we are done as the set on the right is in $\mathcal{M}$ being a countable union of sets from $\mathcal{M}$ which is sufficient to prove measurablity via Exercise 2.4 □

Exercise 2.8. If $f : \mathbb{R} \to \mathbb{R}$ is monotone, then f is Borel measurable.

Solution. trivial. □

Exercise 2.9. Let $f : [0,1] \to [0,1]$ be the Cantor function (§1.5), and let $g(x) = f(x) + x$.

a. g is a bijection from $[0,1]$ to $[0,2]$, and $h = g^{-1}$ is continuous from $[0,2]$ to $[0,1]$.

b. If C is the Cantor set, $m(g(C)) = 1$.

c. By Exercise 29 of Chapter 1, $g(C)$ contains a Lebesgue nonmeasurable set A. Let $B = g^{-1}(A)$. Then B is Lebesgue measurable but not Borel.

d. There exist a Lebesgue measurable function F and a continuous function G on $\mathbb{R}$ such that $F \circ G$ is not Lebesgue measurable.

Solution.

a. It is known that the cantor function is continuous on $[0,1]$ as well as being increasing. This makes $g(x) = f(x) + x$ strictly increasing and continuous, with this we can say using the intermediate value theorem that g is is a continuous bijection from $[0,1]$ to $[0,2]$ and $h = g^{-}1$ is also continuous.

b. As the cantor set is borel and h is continuous, it must be borel measurable. Thus, $h^{-1}(C) = g(C)$ is borel (as well as being lebesgue measurable which is weaker). We can write $m(g([0,1])) = m(g(C)) + m(g([0,1] \setminus C))$ as the sets here are disjoint (the second one is clearly measurable). From the previous parts we see that $m(g(C)) = 2 - m(g([0,1] \setminus C))$. This set is the disjoint union of the middle third intervals removed in the process of constructing the cantor set, on which, by defintion the cantor function is constant. These removed open intervals have measures summing to 1 and as g is a translation on these we see that $m(g([0,1] \setminus C)) = 1$ showing that $m(g(C)) = 1$.

c. As the lebesge measure is complete we see that B being the subset of the cantor set which has measure zero, must be lebesgue measurable too. Now, if B was borel then so would be $h^{-}1(B) = A$ as g^{-1} is continuous, so it is not borel.

d. Consider $\chi_B \circ h$, we see that $(\chi_B \circ h)^{-1}(\{1\}) = h^{-1}(\chi_B^{-1}(\{1\})) = h^{-1}(B) = A$ which is not lebesgue measurable while we had h be continuous (a homeomorphism even) and χ_B lebesgue measurable so we are done as $\{1\}$ is borel.

□

Exercise 2.10. Prove that the following implications are valid iff the measure μ is complete:

a. If f is measurable and $f = g$ μ-a.e. then g is measurable.

b. If f_n is measurable for $n \in \mathbb{N}$ and $f_n \longrightarrow f$ μ-a.e. then f is measurable.

Solution.

a. First assume that μ is complete and that f and g disagree on some $N \in \mathcal{M}$ with $\mu(N) = 0$, then for any measurable E in the range we must have that

$$g^{-1}(E) = (g^{-1}(E) \cap N^c) \cup (g^{-1}(E) \cap N) = (g|_{N^c}^{-1}(E)) \cup (g^{-1}(E) \cap N)$$

$$= (\underbrace{f|_{N^c(\in\mathcal{M})}^{-1}(E)}_{\in\mathcal{M}}) \cup (\underbrace{g^{-1}(E) \cap N}_{\subseteq N \implies \in\mathcal{M}})$$

and thus this set must be in $\mathcal{M}$ itself. For the other direction, suppose $\mu(N) = 0$ for some $N \in \mathcal{M}$ and $K \subseteq N$. Let, $f = \chi_N$ and $g = \chi_K$, we see that $f = g$ on N^c and thus $f = g$ μ-a.e. which makes g measurable and thus $g^{-1}(\{1\}) = K$ must also be measurable making μ complete.

b. For one direction assume that μ is complete. By Exercise 2.3 we know that the set where we have convergence, is measurable so let it be E. Further, a.e. convergence implies that $\mu(X \setminus E) = 0$. Let $g = \lim \chi_E f_n(x)$, we see that g is measurable on E as well as on E^c and is thus measurable on X, being the limit (it always exists) of measurable functions. Suppose, N is the set where $f_n \not\longrightarrow f$, by our hypothesis this is of measure zero. For $x \in N^c$ however $f_n(x) \longrightarrow f(x)$ and thus $f(x) = g(x)$ so $f(x) \neq g(x)$ can only happen on N implying that $f = g$ a.e after which part a proves measurablity of f and we are done. For the other direction, suppose E is a set of measure zero and $K \subseteq E$. Let, $f = \chi_K$ and $f_n = 0$, with this we see that $f_n \to f$ on E^c which shows that $f_n \to f$ a.e. and thus f is measurable so we are done.

□

Exercise 2.11. Suppose that f is a function on $\mathbb{R} \times \mathbb{R}^k$ such that $f(x, \cdot)$ is Borel measurable for each $x \in \mathbb{R}$ and $f(\cdot, y)$ is continuous for each $y \in \mathbb{R}^k$. For $n \in \mathbb{N}$, define f_n as follows. For $i \in \mathbb{Z}$ let $a_i = i/n$, and for $a_i \leq x \leq a_{i+1}$ let

$$f_n(x, y) = \frac{f(a_{i+1}, y)(x - a_i) - f(a_i, y)(x - a_{i+1})}{a_{i+1} - a_i}$$

Then f_n is Borel measurable on $\mathbb{R} \times \mathbb{R}^k$ and $f_n \to f$ pointwise; hence f is Borel measurable on $\mathbb{R} \times \mathbb{R}^k$. Conclude by induction that every function on $\mathbb{R}^n$ that is continuous in each variable separately is Borel measurable.

Solution. We can write $\mathbb{R} \times \mathbb{R}^k$ as a countable union of the borel sets $\{[a_i, a_{i+1}] \times \mathbb{R}^k\}_{i \in \mathbb{Z}}$ and proving borel measurablity of f_n on these suffices to prove borel measurablity globally. It being borel on $[a_i, a_{i+1}] \times \mathbb{R}^k$ follows from these maps being clearly borel

$$(x, y) \mapsto (x - a_i), (x, y) \mapsto (x - a_{i+1}), (x, y) \mapsto f(a_i, y), (x, y) \mapsto f(a_{i+1}, y)$$

as the borel measurability is preserved under products and sums. To prove pointwise convergence just notice that for $(x, y) \in [a_i, a_{i+1}] \times \mathbb{R}^k$, $f_n(x, y)$ is a convex combination of $f(a_i, y)$ and $f(a_{i+1}, y)$ which converge to $f(x, y)$ as $n \to \infty$ by continuity of $f(\cdot, y)$. This shows that if $f(\cdot, y)$ is continuous and $f(x, \cdot)$ borel, for all $(x, y) \in \mathbb{R} \times \mathbb{R}^k$ then f is borel. Now, suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is continuous in each variable, suppose that the proposition holds for $n - 1$, then for all $x \in \mathbb{R}, f(x, \cdot) : y \in \mathbb{R}^{n-1} \mapsto f(x, y)$ is borel because this is continuous in each variable as well and for all $y \in \mathbb{R}^{n-1}, f(\cdot, y) : x \in$

$\mathbb{R} \mapsto f(x, y)$ is continuous too, which implies that f is borel. We can induct on n from the base case of $n = 1$ where the proposition is true due to continuity in each variable coinciding with continuity (which is stronger than borel measurability), hence proving the proposition. □

Exercise 2.12. Prove Proposition 2.20. (See Proposition 0.20, where a special case is proved.)

Solution. If $f^{-1}(\{\infty\})$ is not null then the integral will trivially be infinite and thus it must be a null set. Now, consider the sets $H_n = f^{-1}(1/n, \infty]$ and we find that $\{x : f(x) > 0\} = \cup H_n$. Here, we see that $f \geq n^{-1}\chi_{H_n}$ for all $n \geq 1$ and thus $\int f \geq n^{-1}\mu(H_n)$ forces all the H_n to have finite measure. □

Exercise 2.13. Suppose $\{f_n\} \subset L^+$, $f_n \to f$ pointwise, and $\int f = \lim \int f_n < \infty$. Then $\int_E f = \lim \int_E f_n$ for all $E \in \mathcal{M}$. However, this need not be true if $\int f = \lim \int f_n = \infty$.

Solution. For any $E \in \mathcal{M}$ we can see that $f_n\chi_E \longrightarrow f\chi_E$ pointwise as well and thus, by fatou's lemma and the additivity of the integral we have,

$$\int f\chi_E \leq \liminf \int f_n\chi_E \text{ and } \int f\chi_{X\setminus E} \leq \liminf \int f_n\chi_{X\setminus E}$$

$$\implies \int f = \int f\chi_E + \int f\chi_{X\setminus E} \leq \liminf \int f_n\chi E + \liminf \int f\chi_{X\setminus E}$$

Using the fact that the limit infimum is subadditive we see that the above is

$$\leq \liminf \left(\int f_n\chi_E + f_n\chi_{X\setminus E} \right) = \liminf \int f_n = \int f$$

as all the terms involved in this string of inequalities are finite (this follows from the fact that $\int f < \infty$ and that $\int f_n$ is close to $\int f$ eventually) we must have that,

$$\int f\chi_E = \liminf \int f_n\chi_E$$

Assume wlog $\int f_n$ is always finite, then by the superadditivity of the limit supremum and the equality we developed earlier, we have

$$\limsup \int f_n\chi_E = \limsup \left(\int f_n - \int f_n\chi_{E^c} \right) \leq \limsup \int f_n + \limsup \left(-\int f_n\chi_{E^c} \right)$$

$$= \limsup \int f_n - \liminf \int f_n\chi_{E^c} = \int f - \int f\chi_{E^c} = \int f\chi_E$$

$$\limsup \int f_n\chi_E \leq \int f\chi_E = \liminf f_n\chi_E \implies \lim \int f_n\chi_E = \int f\chi_E$$

For a counterexample we can consider a function on $\mathbb{R}$ given by $f_n = n\chi_{(0,1/n)} + \chi_{[1,\infty)}$ which converges pointwise to $\chi_{[1,\infty)}$ and $\int f = \lim \int f_n = \infty$ but if we focus on $(0, 1)$ then $\lim f_n\chi_{(0,1)} = 1$ whereas $\int f\chi_{(0,1)} = 0$. Here the area escaped to infinity in the unit open interval and hence the contradiction. □

Exercise 2.14. If $f \in L^+$, let $\lambda(E) = \int_E f\, d\mu$ for $E \in \mathcal{M}$. Then λ is a measure on $\mathcal{M}$, and for any $g \in L^+$, $\int g\, d\lambda = \int gf\, d\mu$. (First suppose that g is simple.)

Solution. For any disjoint collection $\{E_n\}_{n\in\mathbb{N}} \subseteq \mathcal{M}$ with union E then by the sigma additivity of the integral we have,

$$\lambda(E) = \int f\chi_E d\mu = \int \sum_{n\in\mathbb{N}} f\chi_{E_n} d\mu = \sum_{n\in\mathbb{N}} \int f\chi_{E_n} d\mu = \sum_{n\in\mathbb{N}} \lambda(E_n)$$

Now, suppose $\phi = \sum a_j \chi_{E_j}$ is a simple function, then we see that

$$\int \phi d\lambda = \sum a_j \lambda(E_j) = \sum a_j \int f\chi_{E_j} d\mu = \int \sum a_j f\chi_{E_j} d\mu = \int f\phi d\mu$$

Now, if $g \in L^+$, we can find a sequence of simple functions $\{\phi_n\}$ that increase to g, for these we also see that $f\phi_n$ iincreases to fg and by the monotone convergence theorem and the fact that λ and μ are defined on the same sigma algebra $\mathcal{M}$ we have that,

$$\int fg d\mu = \int \lim f\phi_n d\mu \longleftarrow \int f\phi_n d\mu = \int \phi_n \lambda \longrightarrow \int \lim \phi_n \lambda = \int g\lambda$$

where we used the the monotone convergence theorem on both sides. □

Exercise 2.15. If $\{f_n\} \subset L^+$, f_n decreases pointwise to f, and $\int f_1 < \infty$, then $\int f = \lim \int f_n$.

Solution. Clearly we have $\int f \leq \lim \int f_n < \infty$. For the other direction, consider the measurable set $E = \{f > 0\}$. Fix $\alpha > 1$ and define $F_n := \{x \in E : f_n(x) \leq \alpha f(x)\}$, then $\{F_n\}$ is an increasing sequence of measurable sets and $\cup F_n = E$. From the definiton we see that $\int_{F_n} f_n \leq \alpha \int_{F_n} f$ and $\int_E f_n - \int_{F_n} f_n = \int_{E\setminus F_n} f_n \leq \int_{E\setminus F_n} f_1$. As $A \mapsto \int_A f_1$ is a measure on $\mathcal{M}$ and $E \setminus F_n$ is a decreasing sequence with $\int_{E\setminus F_1} f_1 \leq \int f_1 < \infty$ we see that

$$\lim_{n\to\infty} \int_{F\setminus F_n} f_1 = \int_{\cap E\setminus F_n} f_1 = \int_{E\setminus E} f_1 = 0$$

this implies that $\lim \int_{F_n} f_n = \lim \int_E f_n$. As $A \mapsto \int_A f$ is also a measure, we have

$$\lim \int_{F_n} f_n = \lim \int_E f_n \implies \lim \int_E f_n \leq \lim \alpha \int_{F_n} f = \alpha \lim \int_{F_n} f = \alpha \int_{\cup F_n} f = \int_E f$$

As this is true for any $\alpha > 1$ we have $\lim \int_E f_n \leq \int_E f$. Now, fix $\alpha > 0$ and consider E^c on which f is zero and thus f_n decreases to 0 here. Let $G_n := \{x \in E^c : f_n(x) \leq \alpha f_1(x)\}$ see that $\{G_n\}$ is an increasing sequence of measurable sets with $\cup G_n = E^c$. It can also be shown that $\lim \int_{G_n} f_n = \lim \int_{E^c} f_n$ using similar arguements as we did when working over E and from the definition we see that $\int_{G_n} f_n \leq \alpha \int_{G_n} f_1 \leq \alpha \int f_1$ and thus, $\lim \int_{E^c} f_n = 0 = \int_{E^c} f$ as $\alpha > 0$ was arbitrary. Combining these two deductions, namely $\lim \int_E f_n \leq \int_E f$ and $\lim \int_{E^c} f_n = 0 = \int_{E^c} f$ we see that $\lim \int f_n \leq \int f$ as the integral is additive. Having shown both directions of the inequality we must have $\int f = \lim \int f_n$.
□

Exercise 2.16. If $f \in L^+$ and $\int f < \infty$, for every $\epsilon > 0$ there exists $E \in \mathcal{M}$ such that $\mu(E) < \infty$ and $\int_E f > (\int f) - \epsilon$.

Solution. From Exercise 2.12 we see that the set $E = \{x : f(x) > 0\}$ is sigma finite so we write it as a disjoint(standard technique) union of countably many sets of finite measure as $E = \cup F_n$, now from the additivity of the integral we have,

$$\int f = \int f\chi_E + \underbrace{\int f\chi_{E^c}}_{=0} = \int_E f = \sum_{n\in\mathbb{N}} \int_{F_n} f = \lim_n \sum_{k=1}^{n} \int_{F_n} f = \lim_n \int_{\cup_{k\leq n} F_k} f$$

Now, for any $\epsilon > 0$ we can find a N such that $\int f < \epsilon + \int_{\cup_{k\leq N} F_k} f$ and as $\mu(\cup_{k\leq N} F_k) = \sum_{k\leq N} \mu(F_k) < \infty$ we are done. □

Exercise 2.17. Assume Fatou's lemma and deduce the monotone convergence theorem from it.

Solution. Fatou's lemma implies that $\lim \int f_n = \liminf \int f_n \geq \int \liminf f_n = \int f$ while the other direction of the inequality is trivial. □

Exercise 2.18. Fatou's lemma remains valid if the hypothesis that $f_n \in L^+$ is replaced by the hypothesis that f_n is measurable and $f_n \geq -g$ where $g \in L^+ \cap L^1$. What is the analogue of Fatou's lemma for nonpositive functions?

Solution. Clearly $f_n + g \in L^+$ so we can apply fatou's lemma on it and get

$$\int g + \liminf \int f_n = \liminf \left(\int f_n + \int g \right) = \liminf \int g + f_n$$

$$\geq \int \liminf(f_n + g) = \int g + \liminf f = \int g + \int \liminf f_n$$

and now we can just cancel out the $\int g$ terms. For nonpositive functions f ($-f \in L^+$) we will have that

$$\limsup \int f_n \leq \int \limsup f_n$$

□

Exercise 2.19. Suppose $\{f_n\} \subset L^1(\mu)$ and $f_n \to f$ uniformly.

a. If $\mu(X) < \infty$, then $f \in L^1(\mu)$ and $\int f_n \to \int f$.

b. If $\mu(X) = \infty$, the conclusions of (a) can fail. (Find examples on $\mathbb{R}$ with Lebesgue measure.)

Solution.

a. This is trivial.

b. consider $f_n = n^{-1}\chi_{[0,n^2]}$ on $\mathbb{R}$ with respect to the lebesgue measure.

□

Exercise 2.20. (A generalized Dominated Convergence Theorem) If $f_n, g_n, f, g \in L^1$, $f_n \to f$ and $g_n \to g$ a.e., $|f_n| \leq g_n$, and $\int g_n \to \int g$, then $\int f_n \to \int f$. (Rework the proof of the dominated convergence theorem.)

Solution. We see that $g_n+f_n, g_n-f_n \in L^1$ and also as $f_n \to f, g_n \to g$ a.e. we can restrict the integrals to a set E where convergence is global such that $\int = \int_E$ (as operators) and we work inside E now without mention. By fatou's lemma

$$\int g + \liminf \int f_n = \liminf \int f_n + \lim \int g_n = \liminf \left(\int f_n + \int g_n \right) = \liminf \int f_n + g_n$$

$$\geq \int \liminf (g_n + f_n) \geq \int \liminf g + \liminf f_n = \int f + g = \int f + \int g$$

cancelling out the $\int g$ terms we get $\liminf \int f_n \geq \int f$ and doing the same with $g_n - f_n$ we get $-\limsup \int f_n = \liminf \int -f_n \geq \int -f = -\int f$ which implies that $\int f \geq \limsup \int f$ and thus finally we have $\limsup \int f_n \leq \int f \leq \liminf \int f_n \implies \lim \int f_n = \int f$. □

Exercise 2.21. Suppose $f_n, f \in L^1$ and $f_n \to f$ a.e. Then $\int |f_n - f| \to 0$ iff $\int |f_n| \to \int |f|$. (Use Exercise 20.)

Solution. For the if direction we can use generalised dominated convergence (Exercise 2.20) on $|f_n - f| \to 0$ a.e. where it is dominated by $|f_n| + |f| \to 2|f|$ a.e. as $\int |f_n| + |f| = \int |f| + \int |f_n| \to 2 \int |f| = \int 2|f|$ and for the only if direction we can proceed by the reverse triangle inequality as follows

$$\left| \int |f_n| - \int |f| \right| \leq \int ||f_n| - |f|| \leq \int |f_n - f| \longrightarrow 0$$

□

Exercise 2.22. Let μ be counting measure on $\mathbb{N}$. Interpret Fatou's lemma and the monotone and dominated convergence theorems as statements about infinite series.

Solution. A sequence of functions $\{f_k\}$ on $\mathbb{N}$ corresponds to a double sequence $a_{k,n} = f_k(n)$. Pointwise convergence $f_k \to f$ a.e. means that for each fixed n, $\lim_{k\to\infty} a_{k,n} = a_n$ (since the only set of measure zero for the counting measure is the empty set). With this we can interpret the three theorems as follows:
Fatou's Lemma: If $f_k \in L^+$, then $\int \liminf f_k \leq \liminf \int f_k$. In series notation: If $a_{k,n} \geq 0$ for all $k, n \in \mathbb{N}$, then

$$\sum_{n=1}^{\infty} \left(\liminf_{k\to\infty} a_{k,n} \right) \leq \liminf_{k\to\infty} \sum_{n=1}^{\infty} a_{k,n}$$

Monotone Convergence Theorem: If $\{f_k\} \subset L^+$, $f_k \leq f_{k+1}$, and $f_k \to f$, then $\lim \int f_k = \int f$. In series notation: If $0 \leq a_{1,n} \leq a_{2,n} \leq \dots$ for all n, and $\lim_{k\to\infty} a_{k,n} = a_n$, then

$$\lim_{k\to\infty} \sum_{n=1}^{\infty} a_{k,n} = \sum_{n=1}^{\infty} a_n$$

Dominated Convergence Theorem: If $f_k \to f$ pointwise, and $|f_k| \leq g$ where $g \in L^1(\mu)$, then $\lim \int f_k = \int f$. In series notation: If $\lim_{k\to\infty} a_{k,n} = a_n$ for each n, and there exists a bounding sequence $\{b_n\}$ such that $|a_{k,n}| \leq b_n$ for all k, n, and $\sum_{n=1}^{\infty} b_n < \infty$, then

$$\lim_{k\to\infty} \sum_{n=1}^{\infty} a_{k,n} = \sum_{n=1}^{\infty} a_n$$

□

Exercise 2.23. Given a bounded function $f : [a, b] \to \mathbb{R}$, let

$$H(x) = \lim_{\delta \to 0} \sup_{|y-x| \leq \delta} f(y), \quad h(x) = \lim_{\delta \to 0} \inf_{|y-x| \leq \delta} f(y).$$

Prove Theorem 2.28b by establishing the following lemmas:

a. $H(x) = h(x)$ iff f is continuous at x.

b. In the notation of the proof of Theorem 2.28a, $H = G$ a.e. and $h = g$ a.e. Hence H and h are Lebesgue measurable, and $\int_{[a,b]} H\, dm = \overline{I}_a^b(f)$ and $\int_{[a,b]} h\, dm = \underline{I}_a^b(f)$.

Solution.

a. This is trivial.

b. If it was continuous outside a null set then we can show riemann integrablity by considering partitions of form $P = \{a_1 < b_1 < \ldots < a_n < b_n\} \cup P_\delta$ where, by our assumptions $m(E = \cup(a_i, b_i)) = \epsilon$ is small such that f is continuous on the compact set $[a, b] \setminus E$ and P_δ is of small enough mesh such as to make $U(f|_{[a,b]\setminus E}, P_\delta) - L(f|_{[a,b]\setminus E}) < \delta$. Now, as f is bounded there is some M such that $|f| \leq M$ then

$$U(f, P) - L(f, P) \leq 2M\epsilon + \delta$$

which can be made arbitrarily small so f must be integrable. Now we will prove the converse. If f is integrable then there exists an increasing sequence of finite partitions P_k such that $\|P_k\| \to 0$ and $\lim \int G_{P_k} dm = \int f = \lim \int g_{P_k} dm$. It was already shown in the book that $g = G$ a.e. when f is riemann integrable so if we show that $H = G$ a.e. and $h = g$ a.e. then we would be done using part (a) as this implies $H = h$ a.e. which is equivalent to continuity a.e. by part (a). We will show that $H = G$ a.e. and the rest will follow similarly. Let K be the set of all endpoints of some interval in P_k for some k, this is a null set and we will show that $H = G$ outside it. Say $x \in [a, b] \setminus K$ is any point then we first show that $H \geq G$. First define $H_\delta(x) := \sup_{|x-y|\leq\delta} f(y)$ then we see that $H = \lim_{\delta\to 0} H_\delta$ by definition and $H_\delta \leq H_{\delta'}$ for all $0 \leq \delta \leq \delta'$. Now, for any $\delta > 0$ we can find some k such that $x \in (a_k, b_k) \subseteq (x - \delta, x + \delta)$ where (a_k, b_k) is an interval in P_k as $\|P_k\| \to 0$. Then we must have that $G_{P_k}(x) \leq H_\delta(x)$ and as G_{P_k} is decreasing we have that $G(x) \leq H_\delta(x)$ and then taking the limit as $\delta \searrow 0$ we have $G(x) \leq H(x)$. To show the other direcrion of the equality we notice that for any P_k with the interval (a_k, b_k) containing x within it we can find some $\delta > 0$ such that $(x - \delta, x + \delta) \subseteq (a_k, b_k)$ and this implies that $G_{P_k}(x) \geq H_\delta(x)$ and as $H_\delta(x)$ is decreasing in delta, taking the limit we see that $G_{P_k}(x) \geq H(x)$ and again takinng the limit as $k \to \infty$ we see that $G \geq H$ so we are done.

□

Exercise 2.24. Let $(X, \mathcal{M}, \mu)$ be a measure space with $\mu(X) < \infty$, and let $(X, \overline{\mathcal{M}}, \overline{\mu})$ be its completion. Suppose $f : X \to \mathbb{R}$ is bounded. Then f is $\overline{\mathcal{M}}$-measurable (and hence in $L^1(\overline{\mu})$) iff there exist sequences $\{\phi_n\}$ and $\{\psi_n\}$ of $\mathcal{M}$-measurable simple functions such that $\phi_n \leq f \leq \psi_n$ and $\int(\psi_n - \phi_n)\, d\mu < n^{-1}$. In this case, $\lim \int \phi_n\, d\mu = \lim \int \psi_n\, d\mu = \int f\, d\overline{\mu}$.

Solution. To show the if direction we can consider the functions $g := \sup_n \phi_n$ and $h := \inf_n \psi_n$then these are $\mathcal{M}$-measurable and we have that $\int g - h < 1/n$ for all n and thus it is zero which makes $h = g$ a.e. and thus $f = g = h$ μ-a.e. Now, If h, g are $\mathcal{M}$ measurable then they are clearly $\overline{\mathcal{M}}$ measurable as well and as $\mathcal{M} \subseteq \overline{\mathcal{M}}$ we see that $\overline{\mu}$-a.e. too so we must have that $h = f = g$ $\overline{\mu}$-a.e as well. which implies that f is $\overline{\mathcal{M}}$ measurable. For the converse, we first deal with the case where $f \geq 0$ consider the following simple functions which are $\overline{\mathcal{M}}$ measurable,

$$\phi_n^* := \sum_{k=0}^{2^{2n}-1} \frac{k}{2^n} \cdot \chi_{f^{-1}([k2^{-n},(k+1)2^{-n}))}$$

$$\psi_n^* := \sum_{k=0}^{2^{2n}-1} \frac{k+1}{2^n} \cdot \chi_{f^{-1}([k2^{-n},(k+1)2^{-n}))}$$

Then, for all large enough n we have that $\psi_n^* \geq f \geq \phi_n^*$ and $0 \leq f - \phi_n^*, \psi_n^* - f \leq 2^{-n}$. As f is bounded we can find some constant M with $f \leq M$. For brevity write

$$\phi_n^* = \sum_{k=1}^{m} a_k \chi_{E_k \cup F_k} \text{ where } E_i \cup F_i \text{ are disjoint } E_i \in \mathcal{M}, F_i \subseteq N_i \in \mathcal{M} \text{ with } \mu(N_i) = 0$$

Then define, $\phi_n := \sum_{k=1}^m a_k \chi_{E_k}$ and we have $\phi_n \leq \phi_n^*$ with equality $\overline{\mu}$ a.e. Again writing ψ_n^* in the exact same notation with $N = \cup N_i \in \mathcal{M}$ we see that

$$\psi_n := \chi_{N^c}\psi_n^* + (M+1)\chi_N = (M+1)\chi_N + \sum_{k=1}^{m} a_k \chi_{E_m \cap N^c}$$

is such that $\psi_n \geq \psi_n^*$ with equality with equality $\overline{\mu}$ a.e. (We use $M+1$ instead of M to guarantee the inequality, we need to do this due to how $\psi_n^* \leq f + 2^{-n} \leq M + 1$ for large enough n). We now have $\mathcal{M}$ measurable simple functions ψ_n, ϕ_n such that $\psi_n \geq f \geq \phi_n$ and also $\psi_n = \psi_n^*$ and $\phi_n = \phi_n^*$ $\overline{\mu}$-a.e. (both together), from this we can say that $\psi_n - f, f - \phi_n \leq 2^{-n}$ together $\overline{\mu}$-a.e. implying $\psi_n - \phi_n \leq 2^{-n+1}$ $\overline{\mu}$-a.e. and thus,

$$\int \psi_n - \phi_n d\overline{\mu} = \int \psi_n - \phi_n d\mu \leq \int 2^{-n+1} d\mu = \frac{\mu(X)}{2^{-n+1}} \longrightarrow 0$$

and we can start from a larger index to have $\int \psi_n - \phi_n < n^{-1}$. In the previous line we assumed the following: if g is some $\mathcal{M}$ measurable function then $\int g d\mu = \int g d\overline{\mu}$, this is easy to prove and we omit the proof. We also see that $\phi_n \leq f \leq \phi_n + 2^{-n}$ $\overline{\mu}$-a.e. for large enough n and thus for large n we have $\int \phi_n d\mu \leq \int f d\overline{\mu} \leq \mu(X) \cdot 2^{-n} + \int \phi_n d\mu$ using the assumption from earlier along the way which implies that $\lim \int \phi_n d\mu = \int f d\overline{\mu}$. The same can be proved for ψ_n similarly so we are done with the proof for $f \geq 0$. Now, for general such f we first split it into its measurable (and bounded in this case positive and negative parts, in usual notation $f = f_+ - f_-$. Let ϕ_n^+, ψ_n^+ be as in the previous case, for f_+ and ϕ_n^-, ψ_n^- be those for f_-, then we see that $\phi_n^+ - \psi_n^- \leq f_+ - f_- \leq \psi_n^+ - \phi_n^-$. As $|\phi_n^+ - \psi_n^- - (f_+ - f_-)| \leq |\phi_n^+ - f_+| + |\psi_n^- - f_-| \leq 2^{-n} + 2^{-n}$ $\overline{\mu}$-a.e for large enough n implying

$$\phi_n^+ - \psi_n^- \leq f_+ - f_- \leq \phi_n^+ - \psi_n^- + 2^{-n+1}$$

and we can get the corresponding bounds for $\psi_n^+ - \phi_n^-$ as well and finish like in the previous case using the $\mathcal{M}$ measurable simple functions functions $\zeta_n := \phi_n^+ - \psi_n^-$ and $\gamma_n := \psi_n^+ - \phi_n^-$ with $\zeta_n \leq f \leq \gamma_n$ and $\gamma_n - \zeta_n \leq 2^{-n+2}$ holding $\overline{\mu}$-a.e.

□

Exercise 2.25. Let $f(x) = x^{-1/2}$ if $0 < x < 1, f(x) = 0$ otherwise. Let $\{r_n\}_1^\infty$ be an enumeration of the rationals, and set $g(x) = \sum_1^\infty 2^{-n} f(x - r_n)$.

a. $g \in L^1(m)$, and in particular $g < \infty$ a.e.

b. g is discontinuous at every point and unbounded on every interval, and it remains so after any modification on a Lebesgue null set.

c. $g^2 < \infty$ a.e., but g^2 is not integrable on any interval.

Solution.

a. It follows from the monotone convergence theorem and the fact that the lebesgue integral and riemann integral agree that $\int g dm = \sum_{n=1}^\infty \int 2^{-n} f(x - r_n) dm = \sum_{n=1}^\infty 2^{-n} \cdot \int_0^1 x^{-1/2} dx = 2$ implying $g \in L^1(m)$ which again forces $\{g \geq \infty\}$ to be null.

b. given any real y, if $r_n < y < 1 + r_n$ then we see that $g(y) > 2^{-n}(y - r_n)^{-1/2}$. Now, given any $x \in \mathbb{R}$ and small enough $\delta > 0$ we can find some r_n such that $x - \delta < r_n < x < x + \delta < r_n + 1$ and thus, for all small enough $\epsilon > 0$ with $r_n < r_n + \epsilon < x$ we see that $g(r_n + \epsilon) > 2^{-n}\epsilon^{-2}$ which can be made arbitrarily large, this shows that g is unbounded in any open neighbourhood around x from whcih the claim follows.

c. As $g < \infty$ a.e. we clearly have $g^2 < \infty$ a.e. as well. With the same setup as the previous part, we see that $g(y)^2 > 4^{-n}(y - r_n)^{-1}$ and thus

$$\int_{x-\delta}^{x+\delta} g^2 \mathrm{d}m \geq \int_{r_n}^{x} \frac{1}{2^n (y - r_n)} \mathrm{d}m(y) = \infty$$

thus, its not integrable on any interval of form $(x - \delta, x + \delta)$ proving the claim after noticing how all intervals have an interval of this form.

□

Exercise 2.26. If $f \in L^1(m)$ and $F(x) = \int_{-\infty}^x f(t)\, dt$, then F is continuous on $\mathbb{R}$.

Solution. First we prove right continuity using the sequential criterion. Say $x_n \searrow x$, then as $A \mapsto \int \chi_A |f|$ is a finite measure on $\mathcal{L}$ we see that

$$\lim |F(x_n) - F(x)| = \lim \left| \int_{[x,x_n]} f \mathrm{d}m \right| \leq \lim \int_{[x,x_n]} |f| \mathrm{d}m = \int_{\cap[x,x_n]} |f| dm = 0$$

as $\cap[x, x_n] = \{x\}$ is a lebesge null set and we follow the convention of $0 \cdot \infty = 0$. Right continuity is proven similarly and hence we have continuity. □

Exercise 2.27. Let $f_n(x) = ae^{-nax} - be^{-nbx}$ where $0 < a < b$.

a. $\sum_1^\infty \int_0^\infty |f_n(x)|\, dx = \infty$.

b. $\sum_1^\infty \int_0^\infty f_n(x)\, dx = 0$.

c. $\sum_1^\infty f_n \in L^1([0, \infty), m)$, and $\int_0^\infty \sum_1^\infty f_n(x)\, dx = \log(b/a)$.

Solution. This is just a bunch of calculations. However it furnishes an example to how how $\int \sum |f_n| \notin L^1$ might result in $\int \sum f_n \neq \sum \int f_n$ even whe both are finite and $\sum f_n \in L^1$ □

Exercise 2.28. Compute the following limits and justify the calculations:

a. $\lim_{n\to\infty} \int_0^\infty (1 + (x/n))^{-n} \sin(x/n)\, dx$.

b. $\lim_{n\to\infty} \int_0^1 (1 + nx^2)(1 + x^2)^{-n}\, dx$.

c. $\lim_{n\to\infty} \int_0^\infty n \sin(x/n)[x(1 + x^2)]^{-1}\, dx$.

d. $\lim_{n\to\infty} \int_a^\infty n(1 + n^2x^2)^{-1}\, dx$. (The answer depends on whether $a > 0$, $a = 0$, or $a < 0$. How does this accord with the various convergence theorems?)

Solution.

a. Let, $f_n(x) = (1 + x/n)^{-n} \sin(x/n)$ then it converges $e^{-x} \cdot \sin(0) = 0$ pointwise and it can be checked that $|f_n| \leq (1 + x/2)^{-2}$ for all $n \geq 2$ and as this is L^1, it allows us to conclude that the limit is 0 by DCT.

b. Let $f_n(x) = (1 + nx^2)/(1 + x^2)^n$ then it converges to $\chi_{\{1\}}$ pointwise and as $|f_n| \leq 1$ in the interval $[0, 1]$ we can apply DCT to conclude that the limit is 0

c. In a similar fashion we see that the integrands are dominated by $(1 + x^2)^{-1}$ and converge pointwise to it as well after which we can conclude with DCT that the limit is $\pi/2$ using basic calculus.

d. If $a > 0$ then we see that the integrands are eventually dominated by $(1 + x^2)^{-1}$ as $1/n + nx^2 \geq 1 + x^2 \iff x^2 > 1/n$ which is always true in our case for large enough n allowing us to apply DCT and conclude that the limit is zero as the integrands converge to 0 pointwise. However, if $a \leq 0$ then the integrands converge to $\infty \cdot \chi_{\{0\}}$ pointwise which integrates to 0 whereas the limit is that of the terms $\pi/2 - \arctan(na)$ which has a strictly positive limit here for the $a \leq 0$ we consider. This shows us that DCT *can't be applied* here, in fact it is easily proven without the DCT that any function which dominates this sequence can not be L^1 by considering the values on intervals of form $[1/n, 1/(n - 1)]$ which is left to the reader as an exercise.

□

Exercise 2.29. Show that $\int_0^\infty x^n e^{-x}\, dx = n!$ by differentiating the equation $\int_0^\infty e^{-tx}\, dx = 1/t$. Similarly, show that $\int_{-\infty}^\infty x^{2n} e^{-x^2}\, dx = (2n)!\sqrt{\pi}/4^n n!$ by differentiating the equation $\int_{-\infty}^\infty e^{-tx^2}\, dx = \sqrt{\pi/t}$ (see Proposition 2.53).

Solution. Let $f : [1, \infty) \times [0, \infty) \to \mathbb{R}$ be defined as $f(t, x) = e^{-tx}$, we see that for all t, x we have

$$\left| \frac{\partial}{\partial t} f(x, t) \right| = xe^{-tx} \leq xe^{-x}$$

which is L^1 allowing us to use Theorem 2.27 (page 56) to deduce that

$$\frac{-1}{t^2} = \frac{d}{dt} \int_0^\infty e^{-tx} \mathrm{d}x = \int_0^\infty \frac{\partial}{\partial t} f(x, t) dx = \int_0^\infty -xe^{-tx} dx$$

repeating this a total of n times with the dominating functions being $x^n e^{-x}$ we get the desired conclusion after setting $t = 1$ at the end. For the second parts we work with t over the same interval but this time x varies over $\mathbb{R}$ with $f(t, x) = e^{-tx^2}$ and following the exact same steps with dominating functions of form $x^{2n}e^{-x^2}$ we arrive at a proof. □

Exercise 2.30. Show that $\lim_{k\to\infty} \int_0^k x^n(1 - k^{-1}x)^k \, dx = n!$.

Solution. The integrands $\chi_{[0,k]}x^n(1-x/k)^k$ are dominated by $x^n e^{-x}$ which is L^1 and also the pointwise limit of the integrands. Applying DCT and using the previosu problem we arrive at the required conclusion. □

Exercise 2.31. Derive the following formulas by expanding part of the integrand into an infinite series and justifying the term-by-term integration. Exercise 29 may be useful. (Note: In (d) and (e), term-by-term integration works, and the resulting series converges, only for $a > 1$, but the formulas as stated are actually valid for all $a > 0$.)

a. For $a > 0$, $\int_0^\infty e^{-x^2} \cos ax \, dx = \sqrt{\pi}e^{-a^2/4}$.

b. For $a > -1$, $\int_0^1 x^a(1 - x)^{-1} \log x \, dx = \sum_1^\infty (a + k)^{-2}$.

c. For $a > 1$, $\int_0^\infty x^{a-1}(e^x - 1)^{-1} \, dx = \Gamma(a)\zeta(a)$, where $\zeta(a) = \sum_1^\infty n^{-a}$.

d. For $a > 1$, $\int_0^\infty e^{-ax}x^{-1} \sin x \, dx = \arctan(a^{-1})$.

e. For $a > 1$, $\int_0^\infty e^{-ax}J_0(x) \, dx = (a^2 + 1)^{-1/2}$, where $J_0(x) = \sum_0^\infty (-1)^n x^{2n}/4^n(n!)^2$ is the Bessel function of order zero.

Solution. All of these are very easily proven by expanding some function into its power series representation and then using DCT (with the dominating function being of form $\sum |f_n|$ for functions of form $\sum f_n$) to switch the summation and integration so we state what we expand for each.

a. expand the cosine term and use the previous problem (we set $t = 1$ at the end, here we require the result without that substitution.)

b. expand the denominator.

c. Rewrite the integrand as

$$\frac{x^{a-1}}{e^x - 1} = \frac{x^{a-1}e^{-x}}{1 - e^{-x}} = \sum_{n=0}^{\infty} x^{a-1}e^{-x}(e^{-x})^n$$

d. expand the sine term and use the previous problem.

e. this is already given as a power series

□

Exercise 2.32. Suppose $\mu(X) < \infty$. If f and g are complex-valued measurable functions on X, define

$$\rho(f, g) = \int \frac{|f - g|}{1 + |f - g|} \, d\mu.$$

Then ρ is a metric on the space of measurable functions if we identify functions that are equal a.e., and $f_n \to f$ with respect to this metric iff $f_n \to f$ in measure.

Solution. For the if part, first notice that $x \mapsto x/(1+x)$ is an increasing map on the nonnegatives and thus we have for any arbitrary $\epsilon > 0$ that,

$$\int_X \frac{|f-f_n|}{1+|f-f_n|}d\mu = \int_{|f-f_n|\geq\epsilon} \frac{|f-f_n|}{1+|f-f_n|}d\mu + \int_{|f-f_n|<\epsilon} \frac{|f-f_n|}{1+|f-f_n|}d\mu$$

$$\leq 1 \cdot \mu(\{|f_n - f| \geq \epsilon\}) + \frac{\epsilon}{1+\epsilon} \cdot \mu(\{|f_n - f| < \epsilon\}) \leq \mu(\{|f_n - f| \geq \epsilon\}) + \epsilon \cdot \mu(X)$$

And as $f_n \to f$ in measure we see that $\lim \rho(f_n, f) \leq \epsilon\mu(X)$ which can be made arbitrarily small proving this direction. For the other direction we similarly have that

$$\rho(f_n, f) \geq \frac{\epsilon}{1+\epsilon} \cdot \mu(\{|f_n - f| \geq \epsilon\})$$

As $\rho(f_n, f) \to 0$ we must have the right side of the inequality tend to 0 proving the claim. □

Exercise 2.33. If $f_n \geq 0$ and $f_n \to f$ in measure, then $\int f \leq \liminf \int f_n$.

Solution. By the definition of the limit infimum we can find a subsequence f_{n_k} such that $\liminf \int f_n = \lim \int f_{n_k}$. As f_n converges in measure to f so does f_{n_k} and we can find a subsequence $f_{n_{k_m}}$ that converges a.e. to f. Now, we see that

$$\liminf_n \int f_n = \lim_k \int f_{n_k} \limsup_k \int f_{n_k} \geq \limsup_m \int f_{n_{k_m}} \geq \liminf_m \int f_{n_{k_m}} \geq \int \liminf f_{n_{k_m}}$$

where we use fatou's lemma for the last inequality. As $f_{n_{k_m}} \to f$ a.e., the last expression is just $\int f$ so we are done. □

Exercise 2.34. Suppose $|f_n| \leq g \in L^1$ and $f_n \to f$ in measure.

a. $\int f = \lim \int f_n$.

b. $f_n \to f$ in L^1.

Solution.

a. By Exercise 1.33 we can use fatou's lemma just like we did in the proof for DCT here as well by considering the sequences $g - f_n, g + f_n$ whice are nonnegative and converge to $g - f, g + f$ respectively in measure.

b. By part (a) and the assumption we clearly see that f_n, f are L^1 and as $|f_n| \to |f|$ in measure too while being dominated by g we can apply part (a) again to see that $\int |f| = \lim \int |f_n|$ which is equivalent to the claim by Exercise 1.21.

enumerate □

Exercise 2.35. $f_n \to f$ in measure iff for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $\mu(\{x : |f_n(x) - f(x)| \geq \epsilon\}) < \epsilon$ for $n \geq N$.

Solution. The only if direction follows directly from the definition and for the if direction, given $\epsilon > 0$ and $\delta > 0$ let, $\nu = \min\{\epsilon, \delta\}$ we see that $\{|f_n - f| \geq \epsilon\} \subseteq \{|f_n - f| \geq \nu\}$ and we can find some $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\mu(\{|f_n - f| \geq \epsilon\}) \leq \mu(\{|f_n - f| \geq \nu\}) < \nu \leq \delta$$

and we are done. □

Exercise 2.36. If $\mu(E_n) < \infty$ for $n \in \mathbb{N}$ and $\chi_{E_n} \to f$ in L^1, then f is (a.e. equal to) the characteristic function of a measurable set.

Solution. As the convergence is L^1 we can find a subsequence $\chi_{E_{n_k}}$ that converges to f outside a set of measure 0. Outside this set we cleary have that f is a charecteristic function and its also measurable restricted to this set being the limit of measurable functions, thus it is a.e. equal to a characteristic function of a measurable set. □

Exercise 2.37. Suppose that f_n and f are measurable complex valued functions and $\phi : \mathbb{C} \to \mathbb{C}$.

a. If ϕ is continuous and $f_n \to f$ a.e., then $\phi \circ f_n \to \phi \circ f$ a.e.

b. If ϕ is uniformly continuous and $f_n \to f$ uniformly, almost uniformly, or in measure, then $\phi \circ f_n \to \phi \circ f$ uniformly, almost uniformly, or in measure, respectively.

c. There are counter examples when the continuity assumptions on ϕ are not satisfied

Solution.

a. This is obvious from the fact that composition by continuous functions conserve pointwise convergence which is clear from the definition of continuity.

b. It is a fact from basic real analysis that composition with uniformly continuous functions conserves uniform convergence and in the case of almost uniform convergence we just restrict ourselves to a suitable set. In the case of convergence in measure, first choose $\delta > 0$ such that $|x - y| < \delta \implies |\phi(x) - \phi(y)| < \epsilon$, then we see that,

$$\{x \in X : |f_n(x) - f(x)| < \delta\} \subseteq \{x \in X : |\phi(f_n(x)) - \phi(f(x))| < \epsilon\}$$

and taking complements,

$$\{x \in X : |f_n(x) - f(x)| \geq \delta\} \supseteq \{x \in X : |\phi(f_n(x)) - \phi(f(x))| \geq \epsilon\}$$

Thus, we see that,

$$0 \longleftarrow \mu(\{x \in X : |f_n(x) - f(x)| \geq \delta\}) \geq \mu(\{x \in X : |\phi(f_n(x)) - \phi(f(x))| \geq \epsilon\})$$

proving this claim.

c. A counterexample for part (a) if continuity of ϕ is not assumed is: $X = \mathbb{R}$ with the lebesgue measure, $\phi(x) = 1 - \chi_{\{0\}}$ and $f_n(x) = 1/n \to 0$. For part (b) in the case of uniform and almost uniform convergence is, in the same measurable space as the previous is: $f_n(x) = x + 1/n \to x$ and $\phi(x) = x^2$, this is also a counterexample for the case of convergence in measure.

□

Exercise 2.38. If $f_n \to f$ in measure and $g_n \to g$ in measure, then $f_n + g_n \to f + g$ in measure, and $f_n g_n \to fg$ in measure if $\mu(X) < \infty$, but not necessarily if $\mu(X) = \infty$.

Solution. The first part follows from the fact that

$$\{x \in X : |f_n + g_n - (f+g)| \geq \epsilon\} \subseteq \{x \in X : |f_n - f| \geq \epsilon\} \cup \{x \in X : |g_n - g| \geq \epsilon\}$$

from the triangle inequality. Forthe second part it suffices to prove that if $\psi_n \to 0$ in measure then $\psi_n \phi \to 0$ in measure for any measurable ϕ as,

$$|f_n g_n - fg| \leq |f_n - f||g_n| + |f||g_n - g| \leq |g||f_n - f| + |g - g_n||f - f_n| + |f||g_n - g|$$

due to the middle term converging to 0 in measure, which follows from the set inclusion below. For any $\epsilon > 0$

$$\{|f_n - f||g_n - g| \geq \epsilon\} \subseteq \{|f_n - f| \geq \sqrt{\epsilon}\} \cup \{|g_n - f| \geq \sqrt{\epsilon}\}$$

After which we can apply the previous result that convergence in measure is preserved unded addition. Now consider $S_n := \{x \in X : n \leq |\phi| < n+1\}$. Then $\{S_n\}_{n \geq 0}$ is a partition of X. We see that

$$\sum_{n \geq 0} \mu(S_n) = \mu(X) < \infty$$

and thus for any $\epsilon > 0$ there exists some m_ϵ such that

$$\sum_{n > m_\epsilon} \mu(S_n) < \epsilon/2$$

Implying that,

$$\mu(\{x \in X : |\phi\psi_k| \geq \epsilon\}) < \sum_{n=0}^{m_\epsilon} \mu(\{x \in S_n : |\psi_k \phi| \geq \epsilon\}) + \epsilon/2$$

From construction, for $k = 0, 1, \ldots, m_\epsilon$ we have that

$$\{x \in S_n : |\phi\psi_k| \geq \epsilon\} \subseteq \{x \in X : |\psi_k| \geq \frac{\epsilon}{n+1}\} \subseteq \{|\psi_k| \geq \frac{\epsilon}{m_\epsilon + 1}\}$$

Let N be such that $k \geq N \implies \mu(\{|\psi_k| \geq \epsilon/(m_\epsilon + 1)\}) < \epsilon/2(m_\epsilon + 1)$. Then from the previous work we see that

$$\mu(\{|\phi\psi_k| \geq \epsilon\}) < \sum_{n=0}^{m_\epsilon} \mu(\{x \in S_n : |\psi_k \phi| \geq \epsilon\}) + \epsilon/2 < \frac{(m_\epsilon + 1)\epsilon}{2(m_\epsilon + 1)} + \frac{\epsilon}{2} = \epsilon$$

for all $k \geq N$ implying convergence in measure (by Exercise 1.35) and we are done. For a counter example in the case of $\mu(X) = \infty$ consider $g_n(x) = x^2$ and $f_n(x) = 1/n$ on $\mathbb{R}$ with the lebesgue measure. □

Exercise 2.39. If $f_n \to f$ almost uniformly, then $f_n \to f$ a.e. and $f_n \to f$ in measure.

Solution. Suppose E_n are measurable sets such that $\mu(E_n^c) < 1/n$ then we have that $\mu((\cup E_n)^c = \cap E_n^c) < 1/n$ for all n and hence 0. If $x \in \cup E_n$ then $x \in E_n$ for some n and thus $f_n(x) \to f(x)$ as we have uniform convergence which implies pointwise convergence on E_n. Thus, we have convergence a.e. due to $(\cup E_n)^c$ being null. Given $\delta > 0$, almost uniform convergence gives a set E with $\mu(E^c) < \delta$ such that $f_n \to f$ uniformly on E. Because of uniform convergence, for large enough n, $|f_n(x) - f(x)| < \epsilon$ for all $x \in E$. This means that the set $\{|f_n - f| \geq \epsilon\}$ is entirely contained within E^c. Therefore, its measure is less than δ. Since δ was arbitrary, the measure tends to 0. □

Exercise 2.40. In Egoroff's theorem, the hypothesis "$\mu(X) < \infty$" can be replaced by "$|f_n| \leq g$ for all n, where $g \in L^1(\mu)$."

Solution. We first restrict ourselves to a set witha null complement such that we have pointwise convergence on it and assume this to be the whole space here on which causes no problems due to the complement being null. It would suffice for the sets $E_1(k)$ to have finite measure, which they do as

$$\bigcup_{m \geq n} \left\{ x \in X : |f_m(x) - f(x)| \geq \frac{1}{k} \right\} \subseteq \left\{ x \in X : 2g(x) \geq \frac{1}{k} \right\}$$

with the set on the right being of finite measure due to g being L^1 (more precisely, it is easily shown that it has measure atmost $2k \int_X g$) $\square$

Exercise 2.41. If μ is σ-finite and $f_n \to f$ a.e., there exist measurable $E_1, E_2, \ldots \subset X$ such that $\mu((\bigcup_1^\infty E_j)^c) = 0$ and $f_n \to f$ uniformly on each E_j.

Solution. Using sigma finiteness write $X = \cup E_n$ with E_n being finite. We can apply egoroff's on each of the E_n's to get $E_n^{(m)} \subseteq E_n$ on which we have uniform convergence, with $\mu(E_n \setminus E_n^{(m)}) < 1/m$, then we see that $\mu(E_n \setminus \cup_m E_n^{(m)}) = 0$. As,

$$X \setminus \bigcup_{m,n} E_n^{(m)} = \left(\bigcup_n E_n \right) \setminus \bigcup_n \bigcup_m E_n^{(m)} \subseteq \bigcup_n \left(E_n \setminus \bigcup_m E_n^{(m)} \right)$$

we see that the collection $\{E_n^{(m)}\}_{m,n}$ satisfies the condition in the problem statement as a countable union of null sets (which are $E_n \setminus \cup_m E_n^{(m)}$ here) is still null. $\square$

Exercise 2.42. Let μ be counting measure on $\mathbb{N}$. Then $f_n \to f$ in measure iff $f_n \to f$ uniformly.

Solution. For the only if direction notice that, for any $\epsilon > 0$ if $\mu(\{|f_n - f| \geq \epsilon\}) < 1/2$ for all $n \geq N$ then this set is empty implying that $|f_n(x) - f(x)| < \epsilon$ for all $x \in \mathbb{N}$, for all $n \geq N$ proving uniform convergence. The only if direction is always true regardless of the domain measurable space as, if $|f_n(x) - f(x)| < \epsilon$ for all n large enough, then $\{|f_n - f| \geq \epsilon\}$ is eventually always empty and thus has measure 0 for large enough n making the measure tend to 0 as $n \to \infty$ proving convergence in measure. $\square$

Exercise 2.43. Suppose that $\mu(X) < \infty$ and $f : X \times [0,1] \to \mathbb{C}$ is a function such that $f(\cdot, y)$ is measurable for each $y \in [0,1]$ and $f(x, \cdot)$ is continuous for each $x \in X$.

a. If $0 < \epsilon, \delta < 1$ then $E_{\epsilon,\delta} = \{x : |f(x,y) - f(x,0)| \leq \epsilon \text{ for all } y < \delta\}$ is measurable.

b. For any $\epsilon > 0$ there is a set $E \subset X$ such that $\mu(E) < \epsilon$ and $f(\cdot, y) \to f(\cdot, 0)$ uniformly on E^c as $y \to 0$.

Solution.

a. By continuity we have that,

$$E_{\epsilon,\delta} = \bigcap_{r \in \mathbb{Q} \cap (0,\delta)} \{x \in X : |f(x,r) - f(x,0)| \leq \epsilon\}$$

as the sets we take the intersection of are measurable, so is the set itself.

b. This is almost like a continuous version (as in, the variable we judge uniform convergence over is continuous) of egoroff's theorem. Consider the sets

$$F_n(k) = \bigcup_{m \geq n} E^c_{1/k,1/m}$$

$$= \left\{ x \in X : \exists m (m \geq n) \text{ such that } |f(x,y) - f(x,0)| \geq \frac{1}{k} \text{ for some } y < \frac{1}{m} \right\}$$

By continuity we know that for any fixed $x \in X$ we have $f(x,y) \to f(x,0)$ as $y \to 0$, thus for any fixed k, we see that $\cap_n F_n(k)$ is empty as if some x were to be in it then for any n we could find some $m \geq n$ such that there exists some y with $0 < y < 1/m \leq 1/n$ and $|f(x,y) - f(x,0)| \geq 1/k$ which would contradict continuity as there is some n such that $y < 1/n$ implies $|f(x,y) - f(x,0)| < 1/k$. As we have $\mu(X) < \infty$ and $F_n(k)$ decreases in n we see that $\lim_n \mu(F_n(k)) = \mu(\cap_n F_n(k)) = 0$. For any $\epsilon > 0$, for all k we can find some n_k with $\mu(F_{n_k}(k)) < \epsilon 2^{-k}$. Thus, $\mu(\cup_k F_{n_k}(k)) < \epsilon$ and as,

$$\left(\bigcup_k F_{n_k}(k) \right)^c = \bigcap_k (F_{n_k}(k))^c$$

$$= \left\{ x \in X : \forall k, m \geq n_k \implies |f(x,y) - f(x,0)| < \frac{1}{k} \text{ for all } y < \frac{1}{m} \right\}$$

Let, $E = \cup_k F_{n_k}(k)$, then for all $x \in E^c$, if $y < 1/n_k$ then $|f(x,y) - f(x,0)| < 1/k$ and thus we have uniform convergence on this set. As $\mu(E) < \epsilon$ we are done.

□

Exercise 2.44. (Lusin's Theorem) If $f : [a,b] \to \mathbb{C}$ is Lebesgue measurable and $\epsilon > 0$, there is a compact set $E \subset [a,b]$ such that $m(E^c) < \epsilon$ and $f|_E$ is continuous. (Use Egoroff's theorem and Theorem 2.26.)

Solution. Fix an $\epsilon > 0$, then following a a similar arguement as we did in Exercise 1.38 we can find some M such that $m(\{x \in [a,b] : |f(x)| \geq M\}) < \epsilon$, let the complement of this set be E. Then, as f is bounded on E, it is L^1 and using Theorem 2.26 we can find continuous functions arbitrarily close to this on E in the L^1 metric. We can hence choose a sequence of continuous functions that converge to f on E in the L^1 metric, due to which we can choose a subsequence of this sequence that converges in measure to our function by Corollary 2.32, after that using Egoroff's theorem we can find some measurable $F \subseteq E$ with $m(E \setminus F) < \epsilon$ on which convergence is uniform. As the limit of an uniformly convergent sequence of continuous functions is continuous we have that f is continuous on F. Now, as $F^c \subseteq (E \setminus F) \cup E^c$ we see thatm $m(F^c) = m(E^c) + m(E \setminus F) < \epsilon + \epsilon = 2\epsilon$. As the lebesgue measure is inner regular (Theorem 1.18 on page 36) we can find some compact $K \subseteq F$ such that $\mu(F \setminus K) < \epsilon$ and thus similarly, $m(K^c) = m(F \setminus K) + m(F^c) < \epsilon + 2\epsilon = 3\epsilon$. Thus, we have found a compact set $K \subseteq [a,b]$ with $m(K^c) < 3\epsilon$ on which f is continuous. □

Exercise 2.45. If $(X_j, \mathcal{M}_j)$ is a measurable space for $j = 1,2,3$, then $\bigotimes_1^3 \mathcal{M}_j = (\mathcal{M}_1 \otimes \mathcal{M}_2) \otimes \mathcal{M}_3$. Moreover, if μ_j is a σ-finite measure on $(X_j, \mathcal{M}_j)$, then $\mu_1 \times \mu_2 \times \mu_3 = (\mu_1 \times \mu_2) \times \mu_3$.

Solution. Having identified $A\times b\times C$ with $(A\times B)\times C$ in the text, we find that by definiton we see that $\otimes_1^3\mathcal{M}_j = (\mathcal{M}_1\otimes\mathcal{M}_2)\otimes\mathcal{M}_3$ as each of these clearly contains the generators of the other. These two measures also clearly agree on the measurable rectangles and thus we can say that they are equal on the algebra formed by the finite disjoint unions of these rectangles, hence we can say that they are equal on the whole sigma algebra by the uniqueness criterion in caratheodory's extension theorem from chapter 1 as the premeasures in this case are clearly σ-finite (the reasoning behind this is given for the case of two measurable spaces at the bottom of page 64 and we thus omit the extremely similar arguement needed here) □

Exercise 2.46. Let $X = Y = [0,1]$, $\mathcal{M} = \mathcal{N} = \mathcal{B}_{[0,1]}$, μ = Lebesgue measure, and ν = counting measure. If $D = \{(x,x) : x \in [0,1]\}$ is the diagonal in $X \times Y$, then $\iint \chi_D \, d\mu \, d\nu$, $\iint \chi_D \, d\nu \, d\mu$, and $\int \chi_D \, d(\mu\times\nu)$ are all unequal. (To compute $\int \chi_D \, d(\mu\times\nu)$, go back to the definition of $\mu\times\nu$.)

Solution. D is clearly measurable and thus $\chi_D \in L^+$. We have that $D_x = \{x\}$ and $D^y = \{y\}$, we use this to compute the iterated integrals combined with the fact that for any E we have $(\chi_E)_x = \chi_{E_x}$ and similarly for y as well.

$$\int\int \chi_D d\mu(x)d\nu(y) = \int\int \chi_{D^y}(x)d\mu(x)d\nu(y) = \int\int \chi_{\{y\}}(x)d\mu(x)d\nu(y) = \int \mu(\{y\})d\nu(y) = 0$$

as $\mu(\{y\})$ is always 0.

$$\int\int \chi_D d\nu(y)d\mu(x) = \int \nu(\{x\})d\mu(x) = \int 1 d\mu(x) = \int_{[0,1]} 1\mu(x) = 1$$

We will now show that $\int \chi_D d(\mu\times\nu) = \mu\times\nu(D) = \infty$. If $D \subseteq \cup I_n$ with $I_n \in \mathcal{A}$ then, we can break these I_n into rectangles and have $D \subset \cup A_m \times B_m = \cup I_n$. Clearly we must have $[0,1] \subseteq \cup(A_m \cap B_m)$ as $(x,x) \in A_m \times B_m$ iff $x \in A_m \cap B_m$. Keeping in mind that $A_m, B_m \in \mathcal{B}_{[0,1]}$ we thus have that $\sum \mu(A_m \cup B_m) \geq 1$ which implies that, for some m we must have $\mu(A_m), \mu(B_m) \geq \mu(A_m \cap B_m) > 0$ but then B_m has more than countably many elements making $\nu(B_m) = \infty$ so $\mu\times\nu(A_m \times B_m) = \mu(A_m)\nu(B_m) = \infty$. Thus, in the definiton of $\mu\times\nu$ the numbers we take an infimum over is the singleton $\{\infty\}$ making the measure of D infinite and thus all the integrals are unequal. □

Exercise 2.47. Let $X = Y$ be an uncountable linearly ordered set such that for each $x \in X$, $\{y \in X : y < x\}$ is countable. (Example: the set of countable ordinals.) Let $\mathcal{M} = \mathcal{N}$ be the σ-algebra of countable or co-countable sets, and let $\mu = \nu$ be defined on $\mathcal{M}$ by $\mu(A) = 0$ if A is countable and $\mu(A) = 1$ if A is co-countable. Let $E = \{(x,y) \in X \times X : y < x\}$. Then E_x and E^y are measurable for all x, y, but $\iint \chi_E \, d\mu \, d\nu$ and $\iint \chi_E \, d\nu \, d\mu$ exist but are not equal. (If one believes in the continuum hypothesis, one can take $X = [0,1]$ [with a nonstandard ordering] and thus obtain a set $E \subset [0,1]^2$ such that E_x is countable and E^y is co-countable [in particular, Borel] for all x, y, but E is not Lebesgue measurable.)

Solution. Clearly $E_x = \{y \in X : y < x\}$ and $E^y = \{x \in X : x > y\}$. By definition we see that E_x is always countable and E^y always uncountable having a countable complement $E_y \cup \{y\}$, so these are always measurable. Thus,

$$\int\int \chi_E d\nu d\mu = \int \nu(E_x)d\mu(x) = \int 0 d\mu(x) = 0$$

$$\int\int \chi_E d\mu d\nu = \int \mu(E^y)\nu(y) = \int 1 d\nu(y) = \nu(Y) = 1 \text{ as } Y \text{ is } \blacksquare\text{able}$$

So these integrals are unequal. □

Exercise 2.48. Let $X = Y = \mathbb{N}, \mathcal{M} = \mathcal{N} = \mathcal{P}(\mathbb{N}), \mu = \nu =$ counting measure. Define $f(m,n) = 1$ if $m = n, f(m,n) = -1$ if $m = n+1$, and $f(m,n) = 0$ otherwise. Then $\int f\, d\mu\, d\nu$ and $\int f\, d\nu\, d\mu$ exist and are unequal.

Solution. Firstly we see that $f(k,k) = 1$ and $f(k,k-1) = 1$. Also, f is measurable (all functions are) as $\mathcal{P}(\mathbb{N}) \otimes \mathcal{P}(\mathbb{N}) = \mathcal{M}(\mathcal{P}(\mathbb{N}) \times \mathcal{P}(\mathbb{N})) = \mathcal{P}(\mathbb{N} \times \mathbb{N})$. Integration with respect to these counting measures is just summation, thus we can write,

$$\int\int f d\mu d\nu = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} f(m,n) = 1 + (1-1) + (1-1) + \ldots = 1$$

$$\int\int f d\nu d\mu = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} f(m,n) = (1-1) + (1-1) + \ldots = 0$$

□

Exercise 2.49. Prove Theorem 2.39 by using Theorem 2.37 and Proposition 2.12 together with the following lemmas.

a. If $E \in \mathcal{M} \otimes \mathcal{N}$ (typo in the book) and $\mu \times \nu(E) = 0$, then $\nu(E_x) = 0$ for a.e. x and $\mu(E^y) = 0$ for a.e. y.

b. If f is $\mathcal{L}$-measurable and $f = 0$ λ-a.e., then f_x and f^y are integrable for a.e. x and y, and $\int f_x\, d\nu = \int f^y\, d\mu = 0$ for a.e. x and y. (Here the completeness of μ and ν is needed.)

Solution. We first prove the lemmas.

a. From the assumptions we have that

$$0 = \mu \times \nu(E) = \int \mu(E^y)\nu(y) = \int \nu(E_x)\mu(x)$$

thus, $\nu(E_x) = 0$ for a.e. x and similarly for y.

b. As f is $\mathcal{L} = \overline{\mathcal{M} \otimes \mathcal{N}}$ measurable, by proposition 2.12, there exists a $\mathcal{M} \otimes \mathcal{N}$ measurable function g such that $f = g$ λ-a.e. which clearly implies that $f = g$ $\mu \times \nu$-a.e. as well. Now, if f is 0 λ a.e. then so is g and thus we also similarly have that $g = 0$ $\mu \times \nu$-a.e. We see that g_x and g^y are $\mathcal{N}$ and $\mathcal{M}$ measurable respectively. Suppose we had $f = g$ outside N with $\mu \times \nu(N) = 0$ then by part(a) we see that $\mu(N^y) = 0$ for a.e. y and $\nu(N_x) = 0$ for a.e x. As, $f_x = g_x$ outside N_x which is null for a.e. x, we have that $f_x = g_x$ a.e. for a.e. x. In the context of the theorem, $\mathcal{N}$ is complete which makes f_x measurable if $f_x = g_x$ a.e. As g is 0 $\mu \times \nu$-a.e. we see that $g \in L^1(\mu \times \nu)$ allowing us to use Tonelli's theorem to say that,

$$0 = \int g\, d(\mu \times \nu) = \int\int g_x(y)\, d\nu(y) d\mu(x) \implies \int g_x\, d\nu = 0 \text{ for a.e. } x$$

Combined with the fact that $f_x = g_x$ a.e. (implying $\int f_x = \int g_x$) for a.e. x we see that $\int f_x \, d\nu = 0$ for a.e. x. and thus f_x is integrable for a.e. x as well. The same can be shown for y as well following the same procedure but with different variables.

Now we prove the theorem. The claims concerning the measurablity of f_x and f^y are evident from the proof of the second lemma. Now we will prove the rest assuming the same setup as used above, proving the other implications of the first case where $f \in L^+(\lambda)$ first. We can clearly choose g such that $g \geq 0$ as well (make it 0 on the null set wher it isn't equal to f, i.e. consider $g\chi_{N^c}$ which is also measurable following the same notation as used above) and thus $g \in L^+(\mu \times \nu)$ making $x \mapsto \int g_x \, d\nu$ measurable by Tonelli's theorem. From the second lemma's proof we see that $f_x = g_x$ a.e for a.e. x making $\int f_x \, d\nu = \int g_x \, d\nu$ a.e. and as $\mathcal{M}$ is complete we must have that $x \mapsto \int f_x \, d\nu$ is measurable, being equal a.e. to a measurable function. Now, as $\int g_x \, d\nu = \int f_x \, d\nu$ a.e we also see that $\int \int g \, d\nu d\mu = \int \int f \, d\nu d\mu$. Using Tonelli's theorem again with the fact that the integrals w.r.t the completion and the underlying measure are equal for functions which are measurable w.r.t the underlying measure (this is easily proven) we find that,

$$\int f \, d\lambda = \int g \, d\lambda = \int g \, d(\mu \times \nu) = \int \int g \, d\nu d\mu = \int f \, d\nu d\mu$$

The other order for integration can be dealt with similarly. Now we prove the rest of the implications for part(b). If $f \in L^1(\lambda)$ then as $f = g$ a.e. we see that,

$$\int |g| \, d(\mu \times \nu) = \int |g| \, d(\lambda) = \int |f| d(\lambda) < \infty$$

and thus, $g \in L^1(\mu \times \nu)$. Tonelli's theorem tells us that $g_x \in L^1(\nu)$ for a.e x and that the a.e. defined function $x \mapsto \int g_x \, d\nu$ is integrable (and also measurable outside a null set, this can be easily proven.) As $\mathcal{N}$ is complete we see that it is measurable (we can ignore it on a null set, if this is hard to understand just email me.) Now, clearly we have that $\int f_x = \int g_x$ for a.e. x making $x \mapsto \int f_x \, d\nu$ a mesurable map. Thus, by Tonelli's theorem and the previous line we can say that

$$\int \int f_x(y) \, d\nu(y) d\mu(x) = \int \int g_x(y) \, d\nu(y) d\mu(x) = \int g \, d(\mu \times \nu) = \int f \, d\lambda$$

so we are done. □

Exercise 2.50. Suppose $(X, \mathcal{M}, \mu)$ is a σ-finite measure space and $f \in L^+(X)$. Let

$$G_f = \{(x, y) \in X \times [0, \infty] : y \leq f(x)\}.$$

Then G_f is $\mathcal{M} \times \mathcal{B}_{\mathbb{R}}$-measurable and $\mu \times m(G_f) = \int f \, d\mu$; the same is also true if the inequality $y \leq f(x)$ in the definition of G_f is replaced by $y < f(x)$. (To show measurability of G_f, note that the map $(x, y) \mapsto f(x) - y$ is the composition of $(x, y) \mapsto (f(x), y)$ and $(z, y) \mapsto z - y$.) This is the definitive statement of the familiar theorem from calculus, "the integral of a function is the area under its graph."

Solution. G_f is measurable as the composition of measurable maps is measurable, as hinted in the problem statement. Now, Theorem 2.36 tells us that, $\mu \times m(G_f) =$

$\int m((G_f)_x\, d\mu(x)$ Now, clearly $(G_f)_x = [0, f(x)]$ which has lebesge measure $f(x)$ so we are done. If the inequality was strict then when $f(x) = 0$ the section would be an empty set, which also has $0 = f(x)$ and for $f(x) > 0$ the one sided open interval $[0, f(x))$ still has lebesgue measure $f(x)$. □

Exercise 2.51. Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be arbitrary measure spaces (not necessarily σ-finite).

a. If $f : X \to \mathbb{C}$ is $\mathcal{M}$-measurable, $g : Y \to \mathbb{C}$ is $\mathcal{N}$-measurable, and $h(x, y) = f(x)g(y)$, then h is $\mathcal{M} \otimes \mathcal{N}$-measurable.

b. If $f \in L^1(\mu)$ and $g \in L^1(\nu)$, then $h \in L^1(\mu \times \nu)$ and $\int h\, d(\mu \times \nu) = (\int f\, d\mu)(\int g\, d\nu)$.

Solution. This is obvious for characteristic functions and we can extend it to simple functions bylinearity. Part (a) is trivial as the compositon of measurable maps is measurable so we prove part (b). By Theorem 2.10 there exists measurable simple functions ϕ_n, ψ_n, which tend to f, g such that $|\phi_n| \nearrow |f|$ and $|\psi_n| \nearrow |g|$. By the monotone convergence theorem we see that,

$$\int |fg| = \lim \int |\psi_n \phi_n| = \lim \left(\int |\psi_n| \int |\phi_n| \right) = \int |f| \int |g| < \infty \implies fg \in L^1(\mu \times \nu)$$

Now, by DCT(we use it on the extreme equalities) we have that,

$$\int fg = \lim \int \phi_n \psi_n = \lim \int \phi_n \int \psi_n = \left(\lim \int \phi_n \right) \left(\lim \int \psi_n \right) = \int f \int g$$

□

Exercise 2.52. The Fubini-Tonelli theorem is valid when $(X, \mathcal{M}, \mu)$ is an arbitrary measure space and Y is a countable set, $\mathcal{N} = \mathcal{P}(Y)$, and ν is counting measure on Y. (Cf. Theorems 2.15 and 2.25.)

Solution. By the countable additivity of the integral for any nonnegative measurable function f we have that,

$$\int_{X \times Y} f\, d\mu(\mu \times \nu) = \sum_{y \in Y} \int_{X \times \{y\}} f(x, y)\, d\mu(x) d\nu(y)$$

Now, any mesaurable subset of $X \times \{y\}$ clearly has form $E \times \{y\}$ where E is $\mathcal{M}$ measurable and we see that $\mu \times \nu(E \times \{y\}) = \mu(E)\nu(\{y\}) = \mu(E)$. From the definiton of the integral it is clear that

$$\int_{X \times \{y\}} f(x, y)\, d\mu(x) d\nu(y) = \int_X f(x, y)\, d\mu(x)$$

The map $y \mapsto \int f(x, y)\, d\mu(x)$ is also trivially measurable as the sigma algebra on Y is the whole power set. As ν is the counting measure we thus have,

$$\int \int f(x, y)\, d\mu(x) d\nu(y) = \sum_{y \in Y} \int f(x, y)\, d\mu(x) = \int f\, d(\mu \times \nu)$$

Again, by the additivity of the integral here we see that

$$\int \int f(x, y) d\nu(y) d\mu(x) = \int \sum_{y \in Y} f(x, y) d\mu(x) = \sum_{y \in Y} \int f(x, y) d\mu(x)$$

which is $\int f\, d(\mu \times \nu)$ as we saw before. We also see that the map $x \mapsto \int f(x,y)\, d\nu(y)$ is measurable as,

$$\int f(x,y)\, d\nu(y) = \sum_{y \in Y} f(x,y)$$

as its the limit of a sequence of measurable functions $f^{y_1} + \ldots + f^{y_n}$ (here we enumerate Y as $\{y_n : n \in \mathbb{N}\}$). We have proven Tonelli's theorem for this case and that of Fubini can be proven from this as in the book, or we can also reenact the same steps while using DCT instead of the additivity of the integral (which was just a corollary of MCT in this case). □

Exercise 2.53. Fill in the details of the proof of Theorem 2.41.

Solution. Fix $\epsilon > 0$. Take a sequence of simple measurable functions ϕ_n such that $|\phi_n| \nearrow |f|$ and by DCT, for large enough n we have $\int |\phi_n - f| < \epsilon$, we write $\phi := \phi_n$. Suppose $\phi = \sum_1^m a_j \chi_{E_j}$ where $a_j \neq 0$ and E_j are disjoint, then we see that

$$|a_j| \mu(E_j) = \int_{E_j} |\phi| \leq \int_{E_j} |f| < \infty$$

Implying that $\mu(E_j) < \infty$. By theorem 2.41c, for each E_j we can find a disjoint(disjoincy is neeeded here by the way) collection $\{R_{j,i}\}_{i=1}^{k_j}$ such that $m(E_j \Delta \cup_i R_{j,i}) < \epsilon$ and each $R_{j,i}$ is a product of intervals, we will call these cubes here on out. Define $\psi := \sum_{i,j} a_j \chi_{R_{j,i}}$, then we see that

$$\int |\phi - \psi| \leq \sum_{j=1}^{m} |a_j| \int \cdot \left| \chi_{E_j} - \sum_{j=1}^{k_j} \chi_{R_{j,i}} \right| \leq \sum_{j=1}^{m} |a_j| \cdot m\left(E_j \Delta \bigcup_{i=1}^{k_j} R_{j,i} \right) < \epsilon \cdot \sum_{j=1}^{m} |a_j|$$

Thus, we have that $\int |f - \psi| \leq \int |f - \phi| + \int |\phi - \psi| < \epsilon \cdot (1 + \sum_1^m |a_j|)$ which can be made arbitrarily small. But the fact that ψ is of the form described in the problem requires proof. It suffices to show that any finite union of cubes can be written as a disjont union of such. We consider cubes whose sides (as defined in the book) are closed intervals, the rest follow similarly. Suppose, $A = \prod[a_j, b_j]$ and $B = \prod[c_j, d_j]$ then we clearly have that $A \cap B = \prod([a_j, b_j] \cap [c_j, d_j])$ which is also a cube (it might be empty but thats a cube too.) We also find that

$$A^c = \bigcup_{f \in \{1,c\}^{\{1,\ldots,n\}}} \prod_{j=1}^{n} [a_j, b_j]^{f(i)}$$

by extreme abuse of notation. Notice that this is a disjoint union. Also, $[a_j, b_j]^c = (-\infty, a_j) \cup (b_j, \infty)$ and this allows us to write any of the sets in the union above as a finite disjoint union of cubes as these sets are of form (after permuting the sides so it isn't as notationally strenous to write)

$$((-\infty, a_1) \cup (b_1, \infty)) \times \ldots \times ((-\infty, a_m) \cup (b_m, \infty)) \times [a_{m+1}, b_{m+1}] \times \ldots [a_n, b_n]$$

which can be written as a disjoint union of cubes following this set theoretic identity

$$\prod_{i \in A} \bigcup_{j \in B} S_{i,j} = \bigcup_{f \in B^A} \prod_{i \in A} S_{i,f(j)}$$

as in our case, for fixed i, all the $S_{i,j}$ are disjoint. This shows that the set of all cubes is an elementary family and thus by Propostion 1.7(page 23) we can say that that the

finite disjoint union of cubes is an algebra. This tells us that we can write the union of cubes $\cup_{i,j} R_{j,i}$ as a disjoint union of cubesa and using this we can write ψ as a simple function where all the characteristic functions are those of cubes which are disjoint so we are done with the first claim. Again, by DCT for large enough m we must have that $\int |\psi \cdot \chi_{[-m,m]^n} - f| < 2 \cdot \epsilon$ starting with small enough $\epsilon > 0$. We clearly see that $\psi \chi_{[-m,m]^n}$ vainshes outside a bounded set and as its defined using charactersitc functions on disjoint cubes, we can approximate it on these cubes by a continuius function as we did in the case of the real line, while ignoring cubes of measure 0, which on the real line are just singletons, this allows us to get $\mathcal{O}(\epsilon^n)$ which can be made arbitrarily small. Then we can use the triangle inequality and conclude that $C_c(\mathbb{R}^n)$ is dense in $L^1(\mathbb{R}^n, \mathcal{L}^n, m)$ in the L^1 metric. □

Exercise 2.54. How much of Theorem 2.44 remains valid if T is not invertible?

Solution. Part (b) is true and we prove it below whereas part (a) is false, that is $f \circ T$ is not neccessarily lebesgue measurable, given f is. As T is not invertible we can find some invertible linear map S such that $ST(\mathbb{R}^n) = \mathbb{R}^k \times \{0\}^{n-k}$ where $k < n$ and this is measurable with measure 0. Thus, $m(S^{-1}ST(\mathbb{R}^n)) = |\det S^{-1}| m(ST(\mathbb{R}^n)) = 0$ showing that $T(\mathbb{R}^n)$ is a lebesgue null set. Now, for any $E \in \mathcal{L}^n$(any set even) we have that $T(E) \subseteq T(\mathbb{R}^n)$ proving part (b) by completeness. Now to disprove part (a), say $n = 2$ and consider the indicator function χ of the lebesgue measurabe set $N \times 0$ where N is not lebesgue measurable on the real lina and let $T : (x, y) \mapsto (x, 0)$ then, $T^{-1}(f^{-1}(0.9, 1.1)) = N \times \mathbb{R}$. Now, if $N \times \mathbb{R}$ was in $\mathcal{L}^2$ then its indicator would be measurable and by Theorem 2.39 we would have that f^y is measurable on $\mathbb{R}$ for a.e. y which is false as these are all indicators of N on $\mathbb{R}$ so the set was not measurable to begin with. □

Exercise 2.55. Let $E = [0, 1] \times [0, 1]$. Investigate the existence and equality of $\int_E f \, dm^2$, $\int_0^1 \int_0^1 f(x, y) \, dx \, dy$, and $\int_0^1 \int_0^1 f(x, y) \, dy \, dx$ for the following f:

a. $f(x, y) = (x^2 - y^2)(x^2 + y^2)^{-2}$.

b. $f(x, y) = (1 - xy)^{-a}$ $(a > 0)$.

c. $f(x, y) = (x - \frac{1}{2})^{-3}$ if $0 < y < |x - \frac{1}{2}|$, $f(x, y) = 0$ otherwise.

Solution.

a. We exclude $\{(0, 0)\}$ which has measure 0 and this does not alter anything we care about. As f is now continuous, using Theorem 2.39 we can work in $\mathcal{B}_{\mathbb{R}^2}$ and the positive and negative parts evaluate to ∞ after some basic calculus paired with fubini's theorem and hence the integral does not exist however the iterated integrals evaluate to $\pi/4$ and $-\pi/4$ respectively.

b. Removing $\{(1, 1)\}$ we can say that this is again continuous and following the same arguements above, revert back to $\mathcal{B}_{\mathbb{R}^n}$. As this is nonegative, all the integrals exist and are equal.

c. Its clearly borel, we follow the same arguements above. The first iterated integral exists and equals 0 whereas the other does not. The integral of the positive and negative parts are both infinite and thus the integral over the product measure does not exist.

□

Exercise 2.56. If f is Lebesgue integrable on $(0,a)$ and $g(x) = \int_x^a t^{-1} f(t)\,dt$, then g is integrable on $(0,a)$ and $\int_0^a g(x)\,dx = \int_0^a f(x)\,dx$.

Solution. Consider the function $H(t,x) := \chi_{t>x} \cdot f(t)t^{-1}$ on $(0,a)^2$. H is clearly measurable w.r.t $m \times m$ and by Tonelli's theorem we have that

$$\int_{(0,a)^2} |H|\,d(m\times m) = \int_0^a \int_0^a |H(t,x)|\,dm(t)\,dm(x)$$

$$= \int_0^a \int_0^t \frac{|f(t)|}{t}\,dm(x)dm(t) = \int_0^a |f(t)|\,dm(t) < \infty$$

Making H integrable. Now, by fubini's theorem we can say that $x \mapsto \int_0^a H(x,t)\,dm(t) = \int_a^x f(t)/t\,dm(t) = g(x)$ is measurable and

$$\int_0^a f(t)\,dm(t) = \int_{(0,a)^s} H\,d(m\times m) = \int_0^a \int_0^a H(t,x)\,dm(t)dm(x) = \int_0^a g(x)\,dm(x)$$

□

Exercise 2.57. Show that $\int_0^\infty e^{-sx}x^{-1}\sin x\,dx = \arctan(s^{-1})$ for $s > 0$ by integrating $e^{-sxy}\sin x$ with respect to x and y. (It may be useful to recall that $\tan\left(\frac{\pi}{2} - \theta\right) = (\tan\theta)^{-1}$. Cf. Exercise 31d.)

Solution. Consider $f(x,y) = se^{-sxy}\sin(x)$ on $(0,\infty)\times(1,\infty)$, this is integrable as, by tonelli's we have

$$\int |f| = \int_0^\infty \int_1^\infty se^{-sxy}|\sin(x)|dydx = \int_0^\infty e^{-sx}|\sin(x)/x|dx \le \int_0^\infty e^{-sx} = s^{-1} < \infty$$

Now using fubini's theorem we see that,

$$\int_0^\infty e^{-sx}x^{-1}\sin(x)dx = \int_0^\infty \int_1^\infty f(x,y)dxdy = \int_1^\infty \int_0^\infty se^{-sxy}\sin(x)dxdy$$

$$= \int_1^\infty \frac{s}{1+(sy)^2}dy = \pi/2 - \arctan(s) = \arctan\left(s^{-1}\right)$$

□

Exercise 2.58. Show that $\int e^{-sx}x^{-1}\sin^2 x\,dx = \frac{1}{4}\log(1+4s^{-2})$ for $s > 0$ by integrating $e^{-sx}\sin 2xy$ with respect to x and y.

Solution. Using the same setup as before and the identity $\sin(x)^2 = (1-\cos(2x))/2$ we just need to evaluate integrals of form $\int e^{ax}\cos(bx)dx$ this time instead of using sine there. □

Exercise 2.59. Let $f(x) = x^{-1}\sin x$.

a. Show that $\int_0^\infty |f(x)|\,dx = \infty$.

b. Show that $\lim_{b\to\infty}\int_0^b f(x)\,dx = \frac{1}{2}\pi$ by integrating $e^{-xy}\sin x$ with respect to x and y. (In view of part (a), some care is needed in passing to the limit as $b\to\infty$.)

Solution.

(a) it can be shown with elementary methods that

$$\int_0^n |\sin(x)/x|dx \gg H_n$$

which suffices.

(b) Consider $f(x,y) = e^{-xy}\sin(x)$. For any $b > 0$ we see that,

$$\int_0^b\int_0^\infty |f(x,y)|dydx \le \int_0^b\int_0^\infty xe^{-xy}dydx = b < \infty$$

so its integrable on the domain used for the integral and by fubini's we have,

$$\int_0^b \frac{\sin x}{x}dx = \int_0^b\int_0^\infty f(x,y)dydx = \int_0^\infty\int_0^b f(x,y)dxdy$$

$$= \lim_{b\to\infty}\int_0^\infty -\underbrace{\frac{e^{-by}\sin(b)y + e^{-by}\cos(b) - 1}{1+y^2}}_{g_b(y)}dy$$

We see that $g_b(y) \longrightarrow (1+y^2)^{-1}$ as $b\to\infty$ and $|g_b(y)| \le 3(1+y^2)^{-1}$ which is integrable allowing us to use DCT (by passing to sequences) to conclude that,

$$\int_0^\infty \frac{\sin x}{x}dx = \int_0^\infty \frac{1}{1+y^2}dy = \frac{\pi}{2}$$

□

Exercise 2.60. $\Gamma(x)\Gamma(y)/\Gamma(x+y) = \int_0^1 t^{x-1}(1-t)^{y-1}\,dt$ for $x, y > 0$. (Recall that Γ was defined in §2.3. Write $\Gamma(x)\Gamma(y)$ as a double integral and use the argument of the exponential as a new variable of integration.)

Solution. Here, $x > y$ denotes the set $\{(x,y)\in\mathbb{R}^2 : x > y\}$. Let $f(x,y) = x^{\alpha-1}y^{\beta-1}e^{-(x+y)}$. Then, by Exercise 2.51 we see that

$$\Gamma(\alpha)\Gamma(\beta) = \int_{(0,\infty)^2} f\,d(m\times m)$$

and by Theorem 2.47, using the invertible linear map $G:(x,y)\mapsto(x-y,y)$ we see that,

$$\int_{G(\{x>y\})=(0,\infty)^2} f\,d(m\times m) = \int_{x>y} f(x-y,y)\cdot\underbrace{|\det G|}_{1}d(m\times m)$$

and by Tonelli's we have,

$$\int_{x>y} f(x-y,y) = \int_0^\infty\int_0^x (x-y)^{\alpha-1}y^{\beta-1}e^{-x}dydx$$

$$= \int_0^\infty \int_0^x \left(1 - \frac{y}{x}\right)^{\alpha-1} \left(\frac{y}{x}\right)^{\beta-1} x^{\alpha+\beta-1} e^{-x} dy dx = \int_0^\infty x^{\alpha+\beta-1} e^{-x} \int_0^1 v^{\beta-1}(1-v)^{\alpha-1} dv$$

by a change of variable $v = y/x$. We thus have that

$$\Gamma(\alpha)\Gamma(\beta) = \Gamma(\alpha+\beta) \int_0^1 t^{\alpha-1}(1-t)^{\beta-1} dt$$

proving the claim. □

Exercise 2.61. If f is continuous on $[0, \infty)$, for $\alpha > 0$ and $x \geq 0$ let

$$I_\alpha f(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t)\, dt.$$

$I_\alpha f$ is called the αth **fractional integral** of f.

a. $I_{\alpha+\beta} f = I_\alpha(I_\beta f)$ for all $\alpha, \beta > 0$. (Use Exercise 60.)

b. If $n \in \mathbb{N}$, $I_n f$ is an nth-order antiderivative of f.

Solution.

a. By expanding we have that,

$$I_\alpha(I_\beta f)(x) = \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_0^x \int_0^t (x-t)^{\alpha-1}(t-y)^{\beta-1} f(y) dy dt$$

Heuristically speaking we integrate over the region $x > t > y > 0$. Using the invertible linear map that switches the coordinates and Theorem 2.47 and tonelli's theorem we see that this is equal to

$$\frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_0^x \int_y^x (x-t)^{\alpha-1}(t-y)^{\beta-1} f(y) dt dy$$

$$= \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_0^x f(y)(x-y)^{\alpha+\beta-1} \int_y^x \left(\frac{x-t}{x-y}\right)^{\alpha-1} \left(\frac{t-y}{x-y}\right)^{\beta-1} \frac{dt}{x-y} dy$$

which is $I_{\alpha+\beta} f(x)$ after a change of variable $v = (t-y)/(x-y)$ and an application of the previous problem.

b. This follows from part (a) by repeatedly applying I_1 which is the antiderivative.

□

Exercise 2.62. The measure σ on S^{n-1} is invariant under rotations.

Solution. Let T be a rotation. Given any borel set $E \subseteq S^{n-1}$, let $E_1 = \Phi^{-1}((0,1] \times E)$ as in the text, then we defined $\sigma(E) = n \cdot m(E_1)$ so it suffices to show that $m(E_1) = m(T(E)_1)$. For any $rx' \in E_1$ in usual notation (same as the text) we find that $T(rx') = rT(x')$ with $T(x') \in S^{n-1}$ as well due to T being a rotation ($\|Tv\| = \|v\|$ for all v), thus we have that $T(E_1) = \{rT(x') : r \in (0,1], x' \in E\} = \Phi^{-1}((0,1] \times T(E_1))$ giving us the equality $T(E_1) = T(E)_1$ and by Theorem 2.44 we have $m(T(E_1)) = |\det T| m(E_1) = m(E_1)$ as $|\det T| = 1$ here. □

Exercise 2.63. The technique used to prove Proposition 2.54 can also be used to integrate any polynomial over S^{n-1}. In fact, suppose $f(x) = \prod_1^n x_j^{\alpha_j}$ ($\alpha_j \in \mathbb{N} \cup \{0\}$) is a monomial. Then $\int f\, d\sigma = 0$ if any α_j is odd, and if all α_j's are even,

$$\int f\, d\sigma = \frac{2\Gamma(\beta_1)\dots\Gamma(\beta_n)}{\Gamma(\beta_1+\dots+\beta_n)} \quad \text{where } \beta_j = \frac{\alpha_j+1}{2}.$$

Solution.

$$\int_{\mathbb{R}^n} f(x)e^{-\|x\|^2}\, dm(x) = \int_0^\infty r^{n-1}\int_{S^{n-1}} f(rx')e^{-r^2}\, d\sigma(x')\, dm(r)$$

notice that $f(rx') = r^{\Sigma\alpha_i}f(x')$, this makes the above equal

$$\int_0^\infty r^{n-1+\Sigma\alpha_i}e^{-r^2}\int_{S^{n-1}} f(x')\, d\sigma(x')\, dm(r) = \int f\, d\sigma \int_0^\infty r^{n-1+\Sigma\alpha_i}e^{-r^2}dr$$

Now, by a change of variable we see that,

$$\int_0^\infty r^{n-1+\Sigma\alpha_i}e^{-r^2}dr = \frac{1}{2}\int_0^\infty r^{\frac{n+\Sigma\alpha_i}{2}-1}e^{-r} = \frac{1}{2}\Gamma\Big(\frac{n+\alpha_1+\ldots+\alpha_n}{2}\Big) = \frac{1}{2}\Gamma(\beta_1+\ldots+\beta_n)$$

And by fubini's theorem, unless some α_j is odd we have,

$$\int_{R^n} f(x)e^{-\|x\|^2}\, dm(x) = \prod_{j=1}^n \int_{-\infty}^\infty x^{\alpha_j}e^{-x_j^2}\, dx_j = \prod_{j=1}^n 2\cdot\frac{1}{2}\Gamma\Big(\frac{\alpha+1}{2}\Big) = \Gamma(\beta_1)\cdots\Gamma(\beta_n)$$

now rearranging these proves the claim in this case. If some α_j are odd then one of the integrals $\int_{-\infty}^\infty x^{\alpha_j}e^{-x_j^2}\, dx_j$ would be 0 making $\int f\, d\sigma = 0$ as $\Gamma(\Sigma\beta_i) > 0$ here, this is clear from the definition of the gamma function. □

Exercise 2.64. For which real values of a and b is $|x|^a|\log|x||^b$ integrable over $\{x \in \mathbb{R}^n : |x| < \frac{1}{2}\}$? Over $\{x \in \mathbb{R}^n : |x| > 2\}$?

Solution. By corollary 2.51 we have, after attaching a suitable characteristic function,

$$\int_{B(0,1/2)} \|x\|^a|\log(\|x\|)|^b\, dm(x) = \sigma(S^{n-1})\int_0^{1/2} r^{a+n-1}|log(r)|^b dr$$

It is now trivial to check which a, b work. In the other case the integral is over $(2,\infty)$. □

Exercise 2.65. Define $G : \mathbb{R}^n \to \mathbb{R}^n$ by $G(r,\phi_1,\dots,\phi_{n-2},\theta) = (x_1,\dots,x_n)$ where

$$\begin{aligned} x_1 &= r\cos\phi_1, \quad x_2 = r\sin\phi_1\cos\phi_2, \quad x_3 = r\sin\phi_1\sin\phi_2\cos\phi_3,\dots,\\ x_{n-1} &= r\sin\phi_1\dots\sin\phi_{n-2}\cos\theta, \quad x_n = r\sin\phi_1\dots\sin\phi_{n-2}\sin\theta. \end{aligned}$$

a. G maps $\mathbb{R}^n$ onto $\mathbb{R}^n$, and $|G(r,\phi_1,\dots,\theta)| = |r|$.

b. $\det D_{(r,\phi_1,\dots,\phi_{n-2},\theta)}G = r^{n-1}\sin^{n-2}\phi_1\sin^{n-3}\phi_2\dots\sin\phi_{n-2}$.

c. Let $\Omega = (0,\infty)\times(0,\pi)^{n-2}\times(0,2\pi)$. Then $G|_\Omega$ is a diffeomorphism and $m(\mathbb{R}^n \setminus G(\Omega)) = 0$.

d. Let $F(\phi_1, \ldots, \phi_{n-2}, \theta) = G(1, \phi_1, \ldots, \phi_{n-2}, \theta)$ and $\Omega' = (0, \pi)^{n-2} \times (0, 2\pi)$. Then $(F|\Omega')^{-1}$ defines a coordinate system on S^{n-1} except on a σ-null set, and the measure σ is given in these coordinates by

$$d\sigma(\phi_1, \ldots, \phi_{n-2}, \theta) = \sin^{n-2}\phi_1 \sin^{n-3}\phi_2 \ldots \sin\phi_{n-2}\, d\phi_1 \ldots d\phi_{n-2}\, d\theta.$$

Solution.

(a) Applying the identity $\sin^2(\theta) + \cos^2(\theta) = 1$ collapses $x_1^2 + \ldots + x_n^2$ to r^2 proving the claim.

(b) Consider the transformations $J(r, \phi_1, \ldots, \phi_{n-2}, \theta) = (r\cos\phi_1, r\sin\phi_1, \ldots, \phi_2, \ldots, \phi_{n-2}, \theta)$ and $G(x_1, w, \phi_2, \ldots, \phi_{n-2}, \theta) = (x_1, G(w, \phi_2, \ldots, \phi_{n-2}, \theta))$, then we see that $G = H \circ J$ and by the multivariate chain rule we must have $DG = DH \cdot DJ$, we can calculate DJ easily and DG by induction on the dimension, this proves the claim.

(c) The variables associated to the m-th dimension are subscripted by m. This is trivial for $n = 2$ as $G_2(\Omega_2) = \mathbb{R}^2 \setminus \{(x, 0) : x \geq 0\}$ and this is also clearly a diffeomorphism. Now, we assume it to be true for $n-1$, that is G_{n-1} is a diffeomorphism and $G_{n-1}(\Omega_{n-1}) = U_{n-1}$ where $m(\mathbb{R}^{n-1} \setminus U_{n-1}) = 0$. We also see that $J_n(\Omega_n) = \mathbb{R} \times (0, \infty) \times (0, \pi)^{n-3} \times (0, 2\pi) = \mathbb{R} \times \Omega_{n-1}$ and this is a diffeomorphism as we only alter the first two coordinates and this corresponds to G_2. By the definition of H_n and the assumption we see that $G_n(\Omega_n) = H_n(J_n(\Omega_n)) = \mathbb{R} \times U_{n-1}$ and this is also a diffeomorphism as the first coordinate is unaltered and the rest correpsonds to G_{n-1}. Thus we have that, $\mathbb{R}^n \setminus G_n(\Omega_n) = \mathbb{R}^n \setminus (\mathbb{R} \times U_{n-1}) = \mathbb{R} \times (\mathbb{R}^{n-1} \setminus U_{n-1})$ and G_n being a composition of diffeomorphisms, is one itself. By our convention of $0 \cdot \infty = 0$ we see that this is of measure 0 (one could also use the fact that this has measure $\int_{\mathbb{R}} m_{n-1}(\mathbb{R}^{n-1} \setminus u_{n-1})\, dm_1 = \int 0\, dm_1 = 0$)

(d) By definition we see that $G_n(r, \omega) = rF_n(\omega)$, and as we had $|G_n(r, \omega)| = r$ we see that $F_n(\omega) = G_n(1, \omega)$ maps Ω_n' into S^{n-1}. As G_n was diffeomorphic on $\Omega_n = (0, \infty) \times \Omega_n'$ and acted injectively we must have F_n act diffeomorphically on Ω_n' as its still C^1 (it shares partial derivatives with G_n clearly.) Thus, $F_n|_{\Omega_n'}^{-1}$ forms a valid coordinate system on $F_n(\Omega_n') \subseteq S^{n-1}$. As $G_n(r, \omega) = rF_n(\omega)$ we see that all non-origin points outside a null set can be written uniquely in that form (radius times direction vector on the sphere) and as the radius can be arbitrary, the non-origin points we can not reach must be unreachable due to their direction not being in $F_n(\Omega_n')$. Hence, $(\mathbb{R}^n \setminus U_n) \setminus \{0\} = \{rx' : r \in (0, \infty), x' \in S^{n-1} \setminus F_n(\Omega_n')\}$. We thus see that, as $\chi_{(\mathbb{R}^n \setminus U_n) \setminus \{0\}}(rx') = \chi_{S^{n-1} \setminus F_n(\Omega_n')}(x')$ and by polar integration, the measure of the $(\mathbb{R}^n \setminus G_n(\Omega_n)) \setminus \{0\}$ is

$$\int_0^\infty \int_{S^n} \chi_{S^{n-1} \setminus F_n(\Omega_n)}(x') r^{n-1}\, d\sigma(x')\, dm(r) = \sigma(S^{n-1} \setminus F_n(\Omega_n)) \int_0^\infty r^{n-1}\, dm(r)$$

and we must have the spherical measure be zero here for the whole thing to be zero which was proved in part (c). Now, we see that for borel sets $E \subseteq S^n$,

$$\int_0^1 \int_{S^{n-1}} \chi_E(x') r^{n-1}\, d\sigma(x') dm(r) = \int_{\mathbb{R}^n} \chi_{E_1}\, dm = \int_{G(\Omega_n)} \chi_{E_1}\, dm$$

$$= \int_{\Omega_n} \underbrace{\chi_{E_1}(G(r,\omega))}_{\chi_{(0,1]}(r)\cdot\chi_E(F(\omega))} |\det D_{(r,\omega)}G|\, dm((r,\omega))$$

$$= \int_0^\infty \chi_{(0,1]}(r) r^{n-1} \int_{\Omega'_n} \chi_E(F(\omega)) \sin^{n-2}\phi_1 \sin^{n-3}\phi_2 \cdots \sin\phi_{n-2}\, dm_{n-1}(\omega) dm(r)$$

where we used all the facts developed above freely. Now, after some simplifications we have

$$\sigma(E) = \int_{F^{-1}|^{-1}_{\Omega'_n}(E)} \sin^{n-2}\phi_1 \sin^{n-3}\phi_2 \cdots \sin\phi_{n-2}\, \mathrm{d}m((\phi_1,\phi_2,\ldots,\theta))$$

so we are done.

□